



Introduction to epidemiology and genomic collaborative surveillance in Ireland

(from the perspective of the national
epidemiological surveillance centre)



Wellcome Bacterial Genomic Surveillance Ireland Course
24 June 2026, Trinity College Dublin

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Outline

- Intro to HPSC and remit
- National surveillance system – CIDR
- Linking Epidemiological data and Genomics/Typing data – with examples
 - National reporting, PH investigation
 - International reporting e.g. ECDC: TESSy/EpiPulse
- New national surveillance system – OCIMS

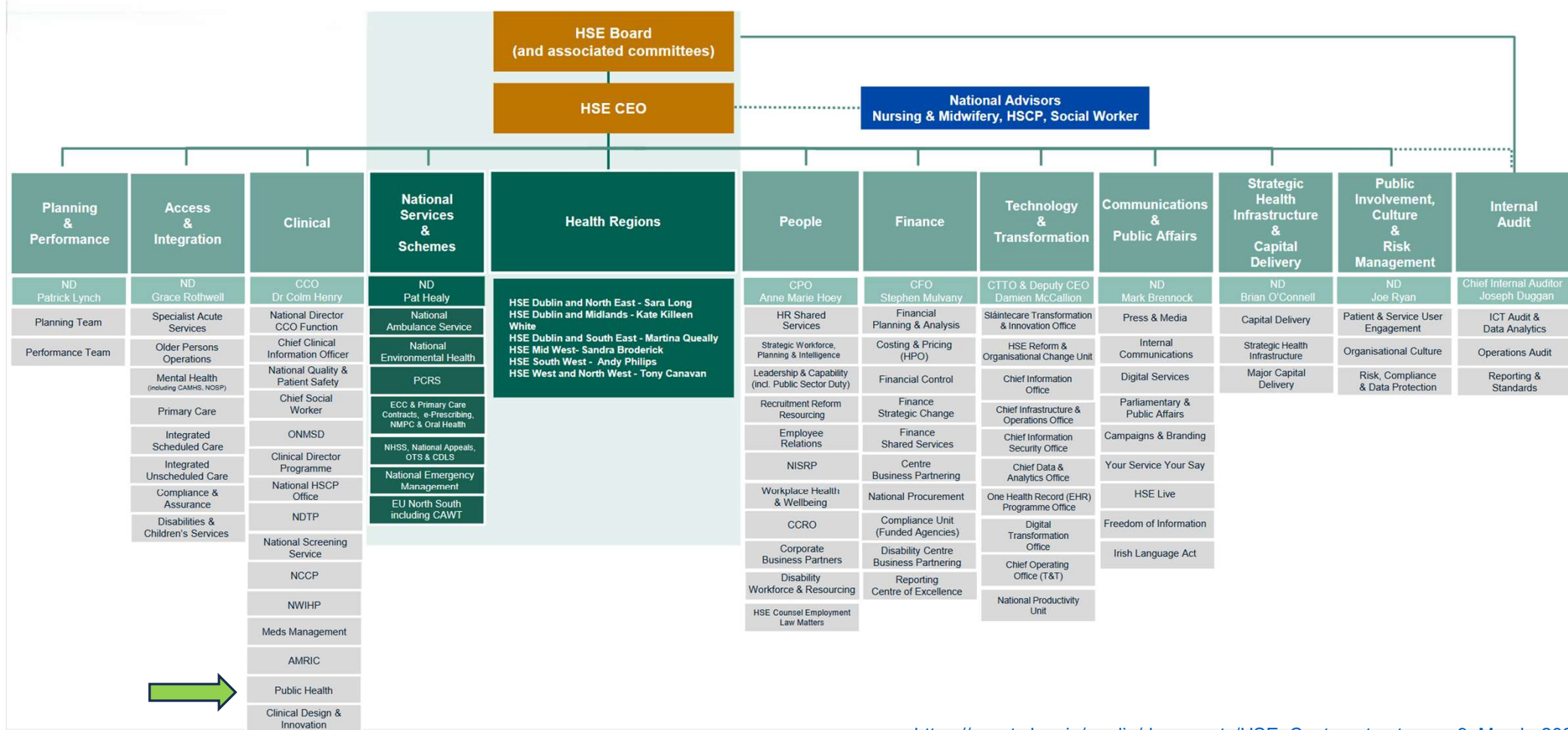


Health Protection Surveillance Centre (HPSC) Overview



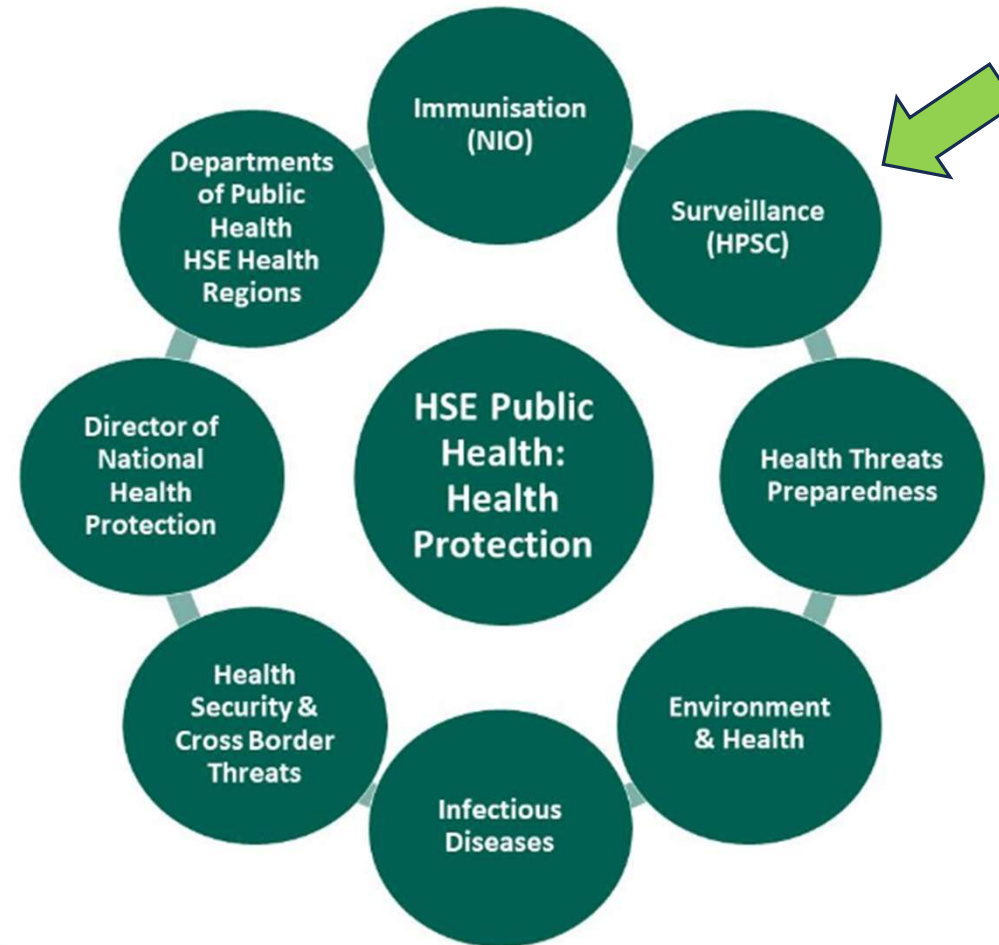


Health Service Executive (HSE) Structure





HPSC within National Health Protection

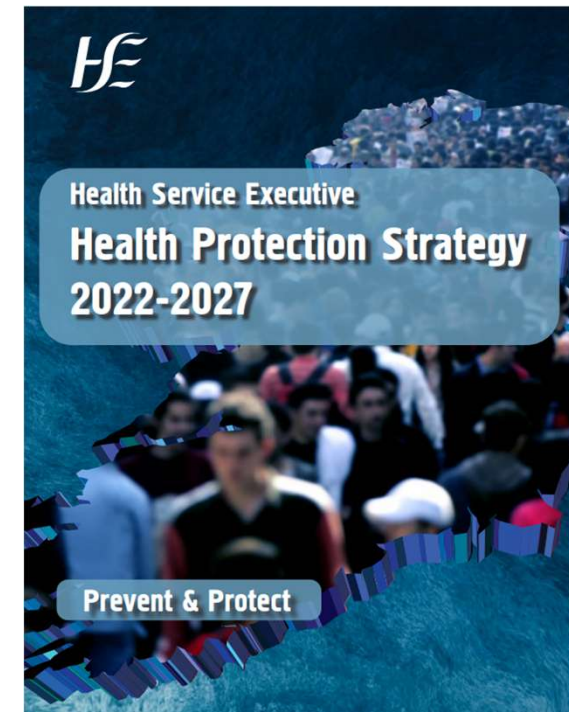
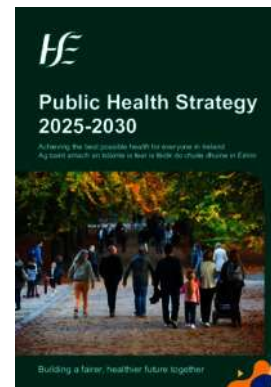
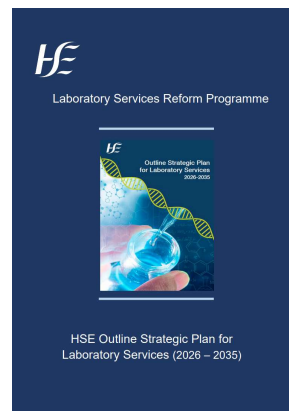




Reform of Health Service, Public Health, Health Protection and Laboratory Services

Structures adapted during the COVID-19 pandemic

- Public Health Reform
- Health Protection Strategy for Ireland 2022–2027
 - All Hazards/One Health approach
- Public Health Strategy 2025–2030
- Laboratory Services Reform Programme 2026–2035





National Public Health Strategy 2025–2030

Enhance integration of public health activities within Public Health, the wider HSE nationally and regionally and drive co-ordination of priorities across sectors, with close collaboration between Regional Executive Officers and Public Health.

Advance Health Intelligence

Leverage health information, evidence, research and innovation for better health

Drive the integration and use of key Public Health datasets including the Outbreak Control Information Management System (OCIMS), National Immunisation Information System (NIIS), Child Health Information System (CHIS), and Laboratory Information Management System (LIMS).

Deliver High-Quality Public Health

Deliver a high-quality and integrated Public Health function

Support a One Health approach, through working in partnerships, to reduce the impact of cross-cutting issues such as emerging zoonotic infections and antimicrobial resistance in diseases impacting human and animal health.

Strengthen Health Services

Strengthen health services to address population needs, emphasising prevention and early intervention

Protect Our Population

Protect the population from all health threats

Leverage existing and new sources of data, including the National Immunisation Information System (NIIS) and Outbreak Case and Incident Management System (OCIMS), in the integrated surveillance, analysis, monitoring, control and prevention of infectious diseases.

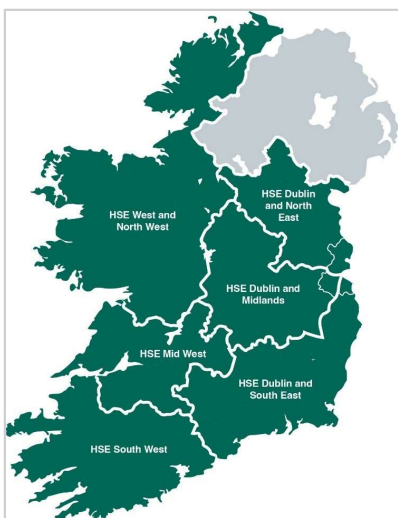


Address the Determinants of Health

Strengthen action on the wider determinants of health

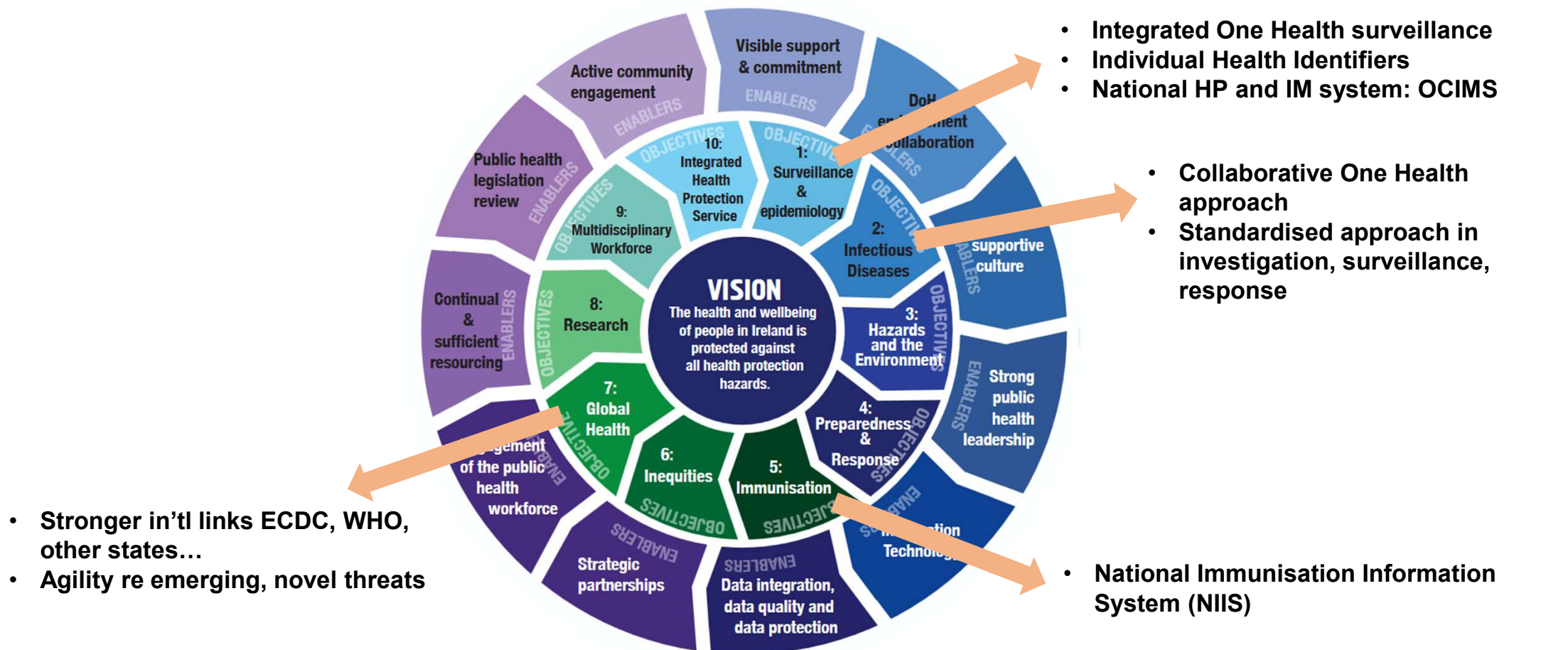
Improve Health Equity

Reduce health inequalities across all stages of life



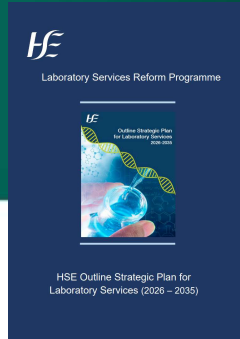


National Health Protection Strategy: 10 Objectives





Laboratory Services Reform Programme: 2026-2035



Key relevant recommendations:

- Development of an integrated Clinical National Reference Laboratory Service
- Development of a HSE Central Laboratory Campus
- Health Protection Laboratory Services should be managed as a single national service
- Development of Laboratory Networks (Hub and Spokes) within the Health Regions
- Inter-laboratory electronic transfer of information in preparation for integration of laboratory systems with the Electronic Health Record (EHR). This will be facilitated by use of the Individual Health Identifier (IHI) Number



HPSC Mission Statement

“To improve the health of the Irish population by provision of the best possible information on disease including infectious diseases through surveillance and independent advice, epidemiological investigation, research and training”

National role of HPSC (ID Regulation S.I. No. 390/1981, Health Acts (1947–2021))

- Surveillance
- Operational support/expert advice
- Training for professionals working in communicable disease control
- Research
- Policy advice
- Public information



• International role of HPSC (National focal point for infectious disease PH under the IHR regulations 2025)

- Accessibility at all times
- Communication with WHO IHR Contact Point
- Dissemination of information to national authorities
- EWRS contact point and competent body for interaction with ECDC



<https://www.hpsc.ie/abouthpsc/>



Notifiable Diseases and their respective causative pathogens

S.I. No. 209/2026 – Infectious Diseases (Amendment) Regulations 2026



90 legally
notifiable
infectious
diseases as
of June
2026

Disease	Causative Pathogen	Disease	Causative Pathogen
Acute anterior poliomyelitis	Polio virus	Lyme Disease (neuroborreliosis)	<i>Borrelia burgdorferi</i>
Ano-genital warts	Human papilloma virus	Lymphogranuloma venereum	<i>Chlamydia trachomatis</i>
Anthrax	<i>Bacillus anthracis</i>	Malaria	<i>Plasmodium falciparum, vivax, knowlesi, ovale, malariae</i>
<i>Bacillus cereus</i> food-borne infection/intoxication	<i>Bacillus cereus</i>	mcr-positive Enterobacteriaceae infection or colonisation	mcr-positive Enterobacteriaceae infection or colonisation
Bacterial meningitis (not otherwise specified)		Measles	Measles virus
Botulism	<i>Clostridium botulinum</i>	Meningococcal disease	<i>Neisseria meningitidis</i>
Brucellosis	<i>Brucella</i> spp.	Mpox	Monkeypox of the Orthopoxvirus genus
<i>Campylobacter</i> infection	<i>Campylobacter</i> spp.	Mumps	Mumps virus
Carbapenemase producing Enterobacteriaceae, infection or colonisation	Carbapenemase producing Enterobacteriaceae, infection or colonisation	Non-specific urethritis	
Chancroid	<i>Haemophilus ducreyi</i>	Novel or Rare Antimicrobial-resistant Organism (NRAO)	
Chickenpox – hospitalised cases	Varicella-zoster virus	Norovirus infection	Norovirus
Chikungunya disease	Chikungunya virus	Paratyphoid	<i>Salmonella</i> Paratyphi
<i>Chlamydia trachomatis</i> infection (genital)	<i>Chlamydia trachomatis</i>	Pertussis	<i>Bordetella pertussis</i>
Cholera	<i>Vibrio cholerae</i>	Plague	<i>Yersinia pestis</i>
<i>Clostridium difficile</i> infection	<i>Clostridium difficile</i>	<i>Pseudomonas aeruginosa</i> infection (invasive)	<i>Pseudomonas aeruginosa</i> (blood or CSF)
<i>Clostridium perfringens</i> (type A) food-borne disease	<i>Clostridium perfringens</i>	Q Fever	<i>Coxiella burnetii</i>
COVID-19	SARS-CoV-2	Rabies	Rabies virus
Creutzfeldt Jakob disease		Respiratory syncytial virus infection	Respiratory syncytial virus
variant Creutzfeldt Jakob disease		Rotavirus infection	Rotavirus
Cryptosporidiosis	<i>Cryptosporidium parvum, hominis</i>	Rubella	Rubella virus
Cytomegalovirus infection (congenital)	Cytomegalovirus	Salmonellosis	<i>Salmonella</i> spp. other than S. Typhi and S. Paratyphi
Dengue fever	Dengue virus	Severe Acute Respiratory Syndrome (SARS)	SARS-associated coronavirus
Diphtheria	<i>Corynebacterium diphtheriae, C. ulcerans</i> or <i>C. pseudotuberculosis</i> (toxin producing)	Shigellosis	<i>Shigella</i> spp.
Echinococcosis	<i>Echinococcus</i> spp.	Smallpox	Variola virus
Enterococcal bacteraemia	<i>Enterococcus</i> spp. (blood)	Staphylococcal food poisoning	Enterotoxigenic <i>Staphylococcus aureus</i>
<i>Escherichia coli</i> infection (invasive)	<i>Escherichia coli</i> (blood, CSF)	<i>Staphylococcus aureus</i> bacteraemia	<i>Staphylococcus aureus</i> (blood)
Giardiasis	<i>Giardia lamblia</i>	<i>Streptococcus</i> group A infection (invasive)	<i>Streptococcus pyogenes</i> (blood, CSF or other normally sterile site)
Gonorrhoea	<i>Neisseria gonorrhoeae</i>	<i>Streptococcus</i> group B infection (invasive)	<i>Streptococcus agalactiae</i> (blood, CSF or other normally sterile site)
Granuloma inguinale	<i>Klebsiella granulomatis</i>	<i>Streptococcus pneumoniae</i> infection (invasive)	<i>Streptococcus pneumoniae</i> (blood, CSF or other normally sterile site)
<i>Haemophilus influenzae</i> disease (invasive)	<i>Haemophilus influenzae</i> (blood, CSF or other normally sterile site)	Syphilis	<i>Treponema pallidum</i>
Hantavirus disease	Hantavirus	Tetanus	<i>Clostridium tetani</i>
Hepatitis A (acute) infection	Hepatitis A virus	Toxoplasmosis	<i>Toxoplasma gondii</i>
Hepatitis B (acute and chronic) infection	Hepatitis B virus	Trichinosis	<i>Trichinella</i> spp.
Hepatitis C infection	Hepatitis C virus	Trichomoniasis	<i>Trichomonas vaginalis</i>
Hepatitis E infection	Hepatitis E virus	Tuberculosis	<i>Mycobacterium tuberculosis</i> complex
Herpes simplex (genital)	Herpes simplex virus	Tularemia	<i>Francisella tularensis</i>
Herpes simplex (neonatal)	Herpes simplex virus	Typhoid	<i>Salmonella</i> Typhi
Human immunodeficiency virus infection	Human immunodeficiency virus	Typhus	<i>Rickettsia prowazekii</i>
Human papilloma virus (HPV) infection (residual unlinked clinical specimens)	Human Papillomavirus (high risk genotypes)	Verotoxigenic <i>Escherichia coli</i> infection	Verotoxin producing <i>Escherichia coli</i>
Influenza	Influenza A and B virus	Viral encephalitis	
<i>Klebsiella pneumoniae</i> infection (invasive)	<i>Klebsiella pneumoniae</i> (blood or CSF)	Viral haemorrhagic fevers	
Legionellosis	<i>Legionella</i> spp.	Viral meningitis	
Leprosy	<i>Mycobacterium leprae</i>	West Nile fever	West Nile virus
Leptospirosis	<i>Leptospira</i> spp.	Yellow fever	Yellow fever virus
Listeriosis	<i>Listeria monocytogenes</i>	Yersiniosis	<i>Yersinia enterocolitica, Yersinia pseudotuberculosis</i>
		Zika virus infection	Zika virus

<https://www.hpsc.ie/notifiablediseases/listofnotifiablediseases/Notifiable%20diseases%20and%20their%20respective%20causative%20pathogens.pdf>
<https://notifiabledisease.hpsc.ie/>



HPSC Unit Structure

Disease Surveillance Units

Gastro, Zoonotic and Vector Borne Disease

VPD and Immunisation Uptake

Respiratory Virus

BBVs-STIs and TB

Antimicrobial Resistance and Healthcare Associated Infections (ARHAI)

Other Surveillance Units

Sentinel and Syndromic Surveillance

Environment and Health

Wastewater, Sero Epidemiology and Legionella

Technology and Analysis

Data Visualisation

Robotic Process Automation

Surveillance Information Systems

Epidemiology Informatics and Digital Process Improvement Team

Biostatistics and Modelling



Computerised Infectious Disease Reporting (CIDR)





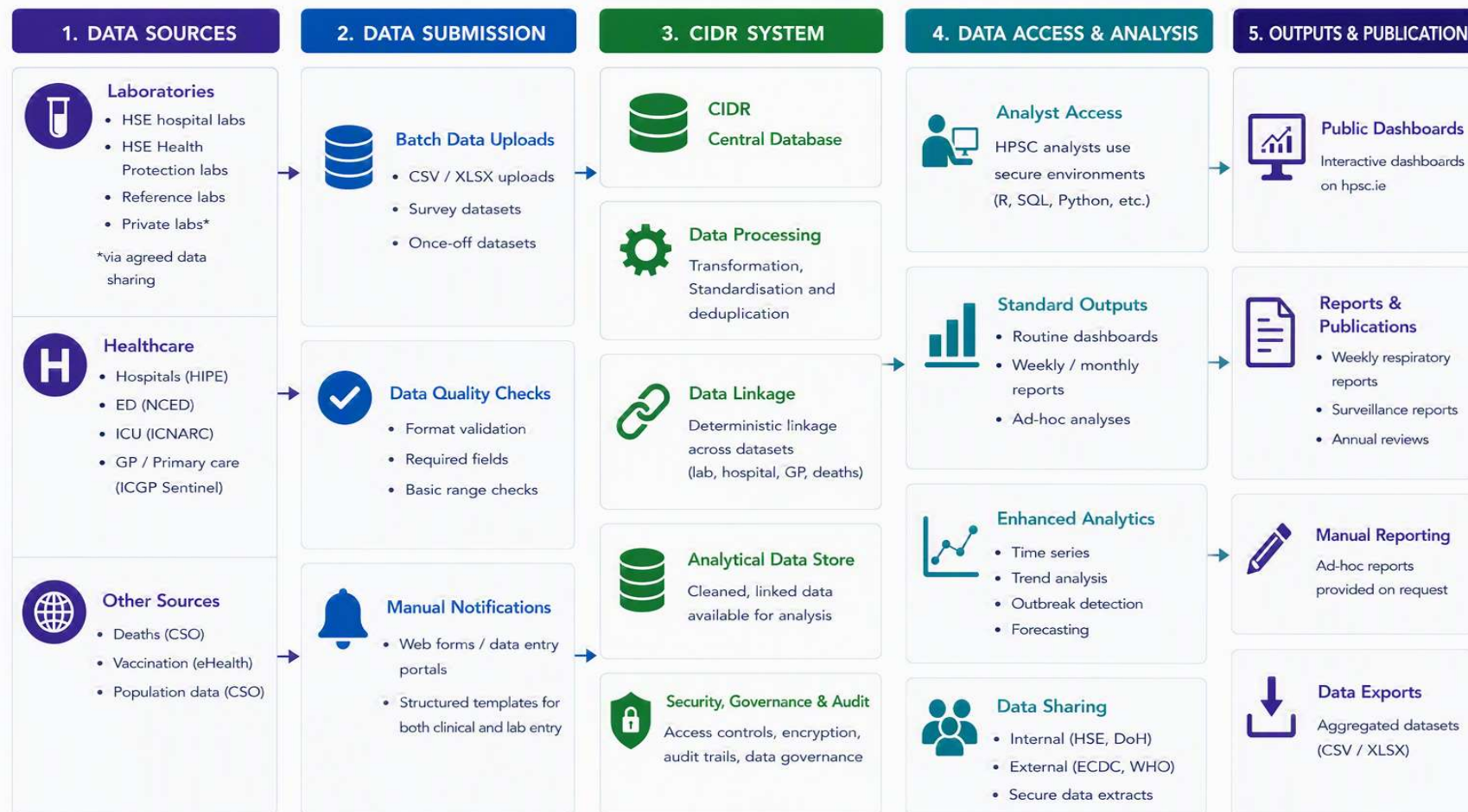
Computerised Infectious Disease Reporting (CIDR)



- Established 2004
- Information system developed to manage the surveillance and control of infectious diseases in Ireland
- Collaborative solution which HPSC manage on behalf of programme stakeholders: Public Health Areas, Food Safety Authority of Ireland, the Food Safety Promotion Board and the Department of Health
- Legal framework: Infectious Diseases Regulations, 1981 and amendments
- Currently, 90 notifiable infectious diseases (IDs), latest S.I. 14th May 2026, 81 on CIDR



CIDR Data Flow





CIDR Limitations

- Coming to End of Life
- Fragmented systems (epi vs lab/genomics not unified)
- Reliance on manual linkage and interpretation
- Limited ability for:
 - Routine, systematic integration across datasets
 - Timely, automated analysis and reporting
 - Constraints on scalability and real-time surveillance



Linking of genomics/typing and epidemiological data



Tuberculosis



- Irish Mycobacteria Reference Laboratory (IMRL) based at St James's Hospital, Dublin (linked with TCD). (MRSA and Gonococcal Reference Labs in same Dept)
- ISO15189 accredited whole genome sequencing service for all *Mycobacterium tuberculosis* complex (MTC) isolates recovered in Ireland
- WGS used primarily for surveillance and cluster/outbreak investigation
 - Drug resistance prediction also available. Confirm phenotypic DST
 - Between 1998 and 2018, selected cases and Mycobacterial Interspersed Repetitive Unit Variable Number Tandem Repeat (MIRU-VNTR) clusters of interest were sequenced (n >600)
 - Transition from MIRU-VNTR to WGS 2019
 - Since 2019, all MTC isolates have been sequenced (n >2,000)
 - IMRL sequenced isolates recovery dates range from 1998 to present day
 - Case data submitted to CIDR with Lineage information





Tuberculosis cluster investigation

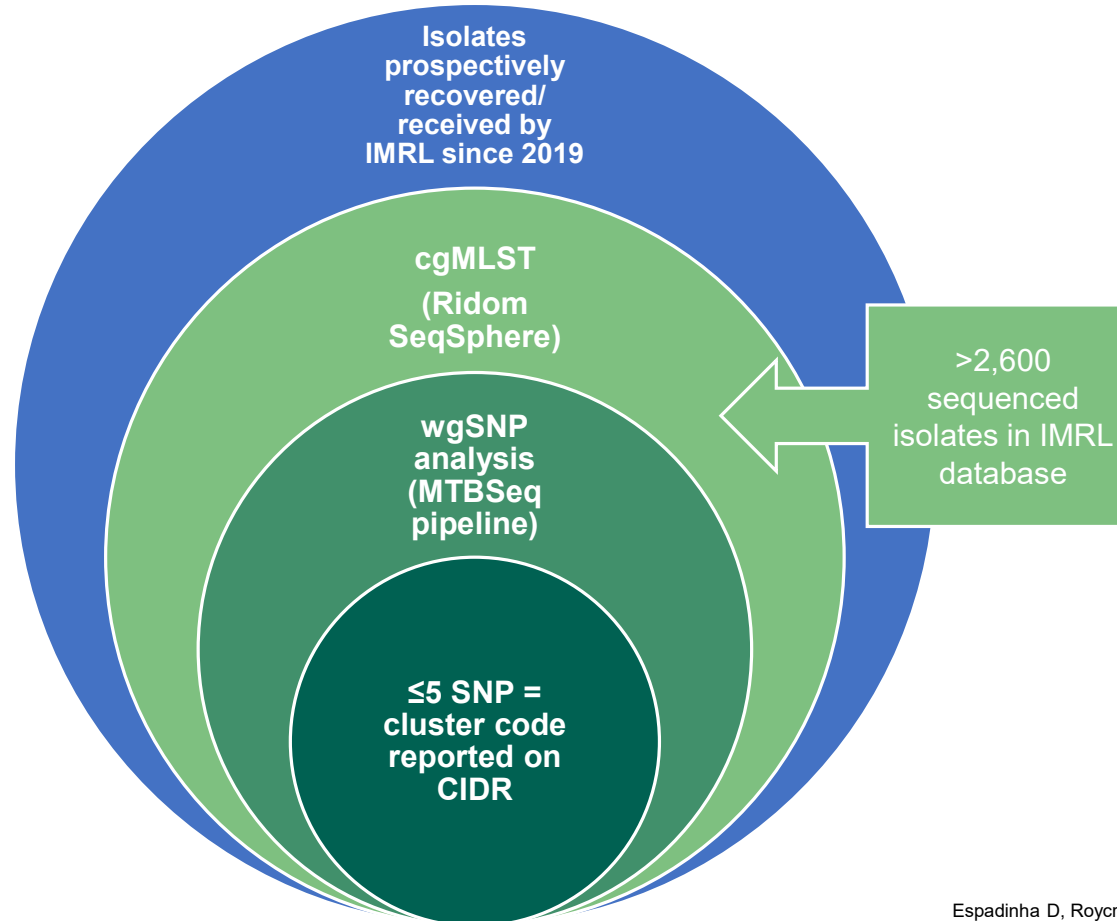


- All clustered isolates from 2011 to present flagged with cluster ID and genotype on CIDR by IMRL. Cluster threshold (≤ 5 SNPs) based on consensus of MTBC community
- TB enhanced surveillance forms completed by PH doctors in the regional PH depts for each TB notification per legislation
 - Forms summarise all available additional clinical and epidemiological data. Data are entered onto CIDR by regional PH depts
 - Fully integrated view of clinical, laboratory and epidemiological data for clustered cases
- Regional Public Health Departments can access lineage/cluster information on CIDR
 - Can run national pseudonymised reports using Cluster ID
 - Prompt further investigation e.g. demographic and epi links > PH intervention as needed
- If epidemiological links detected among clustered cases, outbreak also reported on CIDR





WGS cluster surveillance process



Espadinha D, Roycroft E, Flanagan PR, Mok S, McNamara E, Rogers TR, Fitzgibbon MM, *Validation and implementation of whole-genome sequencing-based analytical methods for molecular surveillance and relatedness analysis of Mycobacterium tuberculosis complex isolates at a national reference laboratory*, Heliyon, 2024, <https://doi.org/10.1016/j.heliyon.2024.e40279>



WGS cluster surveillance, 2011-2025



Number of clusters

- 119 clusters, 12% of total TB cases 2011-2025
- % clustered range: 5% in 2011 to 21% in 2023
- 9 clusters were newly detected in 2025

Cluster size

- Median cluster size 2 cases, mean 4 cases
- Range from 2 - 40 cases

Cluster spread

- 10 clusters span ≥ 3 RHAs
- Longest duration of cluster spans 1998 - 2025



https://www.hpsc.ie/a-z/vaccinepreventable/tuberculosis/tbdataandreports/20260318_2016-2025_TB_Trends_Slideset_v1.1.pdf



TB cluster summary 2025



- 54 cases notified in 2025 linked to 33 clusters identified through WGS by IMRL
- Of these 33 WGS clusters, 9 were confined to 2025 cases
 - cluster start dates ranged from 1998 – 2025
- Most common lineages among clustered cases were Haarlem (n=15) and Euro-American (n=10)
- 10 WGS clusters (30%) were reported as outbreaks:
 - 9 outbreaks represented the entire cluster
 - 1 outbreak represented a subset of epidemiologically linked cases within the WGS cluster





TB national and international reporting



- National Reporting
 - Notifications to CIDR
 - [HPSC Notifiable Disease Hub](#) (high-level weekly updates)
 - [HPSC Outbreaks Hub](#) (high-level weekly updates)
 - [TB Annual Reports](#)
- Reporting to ECDC via EpiPulse platform
 - HPSC report yearly/2x per year upload of Irish data including ENA accession # to ECDC
 - Ongoing IMRL upload of raw reads to ENA
 - ECDC link epi and sequence data (from ENA) for:
 - Cluster analysis; Cross-border clusters reported on EpiPulse. Countries involved can view own data and use Molecular Typing Tool (Microreact)
 - Surveillance; [reports](#), [ECDC Surveillance Atlas](#)





TB Prison Outbreak Investigation: combining epidemiology and genomics



Outbreak

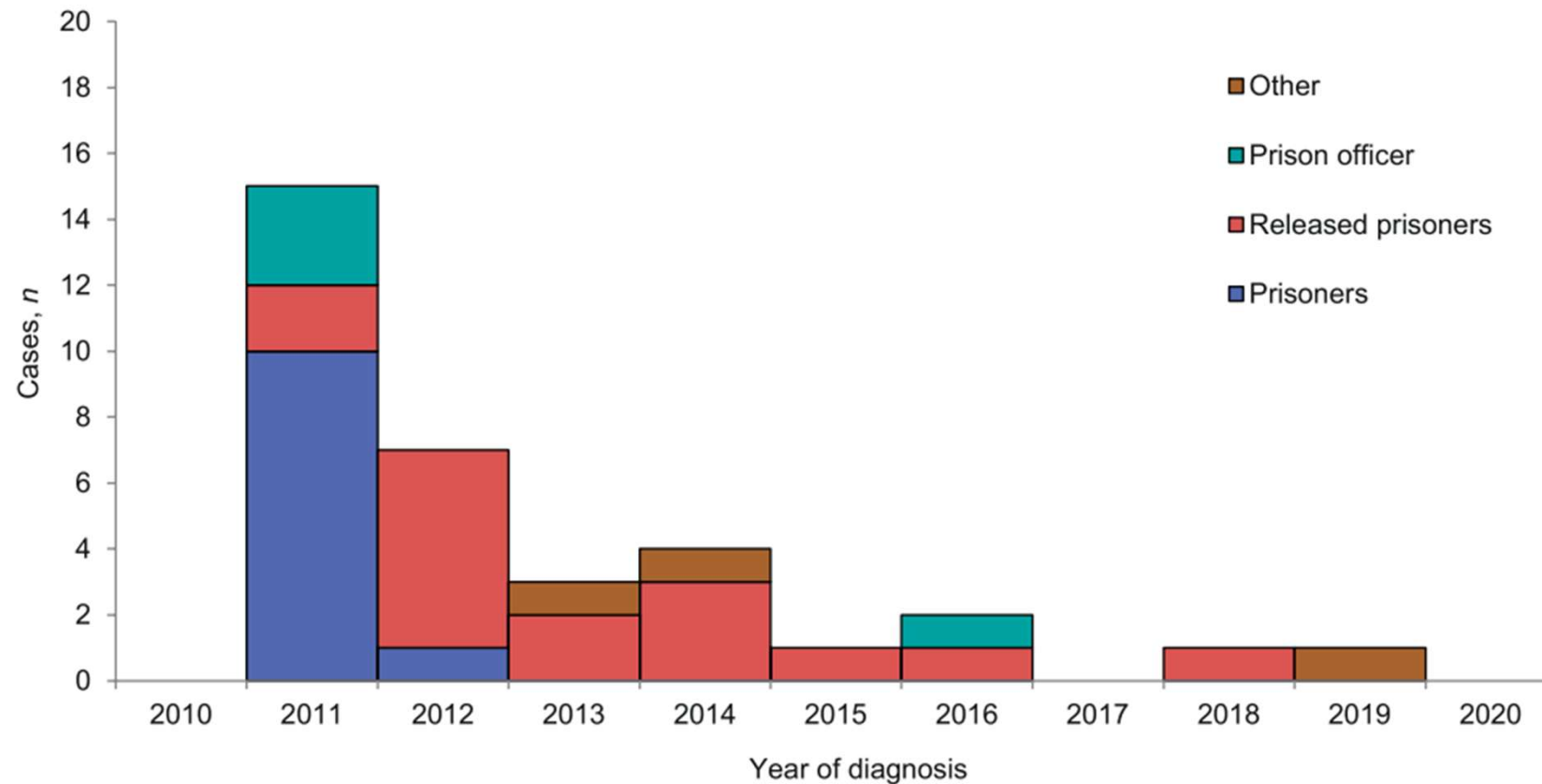
- 2011 DPH East notified outbreak of 2 cases in prisoners sharing cell
- OCT convened: contact tracing of close contacts on same wing and workmates in/outside prison and prison officers
- Further case in separate wing extended investigation
- Further cases identified through active case-finding and enhanced clinical surveillance and screening
- 34 cases between March 2011 to March 2019 but peak of cases in 2011-2012 amongst prisoners/staff/links
- 12 of these found through outreach and enhanced surveillance by Public Health and IMRL from 2013 to 2020

<https://pubmed.ncbi.nlm.nih.gov/34049612/>





TB Prison Outbreak Epi Curve

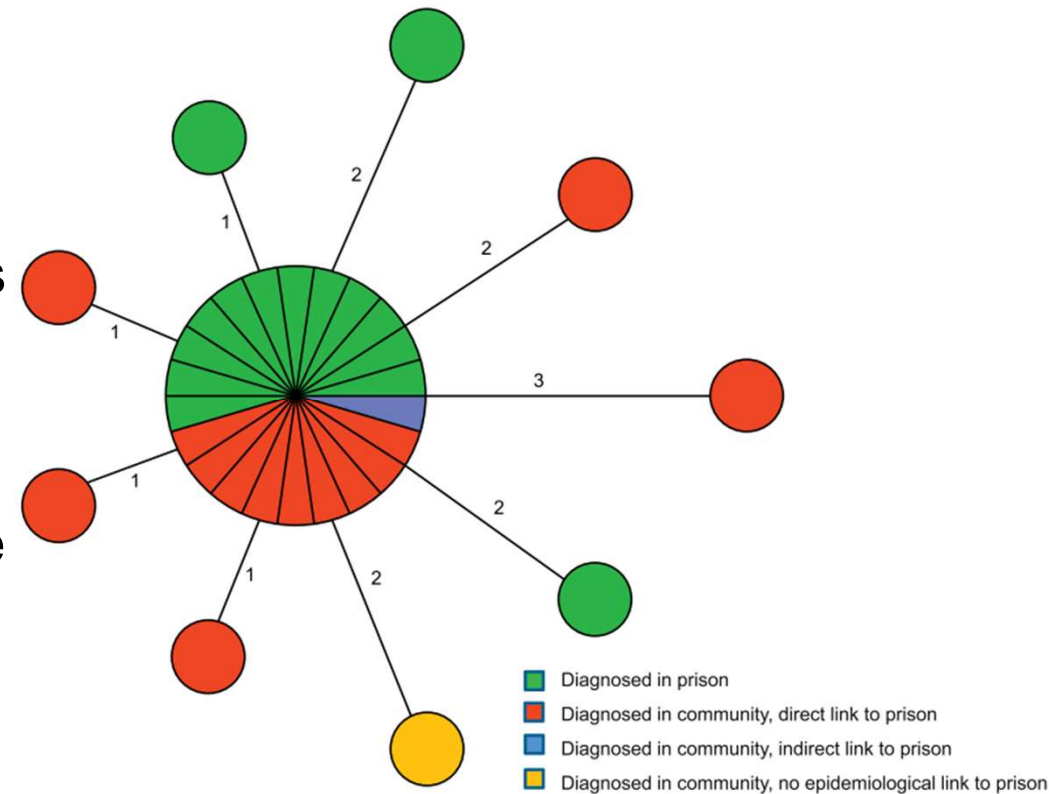


<https://pubmed.ncbi.nlm.nih.gov/34049612/>



Genomic Epidemiology:

- 31 isolates indistinguishable by MIRU-VNTR
- Minimum spanning tree based on SNPs (using Bionumerics) n=31 isolate genomes - Beijing lineage
- Majority of strains within 3 SNPs distant
- 6 SNPs max distance amongst isolates
- Close relationship of most isolates indicate recent transmission and possible 'superspreader'
- Supported by early peak in congregate prison setting and epi investigation



<https://pubmed.ncbi.nlm.nih.gov/34049612/>



Meningococcal disease



- Irish Meningitis and Sepsis Reference Laboratory, Temple St Children's Hospital, Dublin
- WGS information for clinical case management and outbreak investigation
- WGS applications:
 - Lineage determination ...hyperinvasive clone, emerging strains, trends
 - MenY: 17 notified cases, WGS (n=10 isolates received) ST-23 (n=6) or ST-14319 (n=4). Clonal complex 23 associated with disease older adults (IMSRL Annual Report 2022/2023)
 - MenB Vaccine antigen determination (coverage)
 - 22 MenB isolates typed: n=12 cc41/44, PorA VR2 P1.4 epitope gene absent in 3; n=10 cc213 - not anticipated to be recognised by 4CMenB vaccine > possible cases of vaccine failure
 - Fine-scale outbreak investigation, transmission dynamics (especially when paired with epi data)





Meningococcal disease



National Reporting

- Notifications to CIDR
- [HPSC Notifiable Disease Hub](#) (high-level weekly updates)
- [IMD National Reports](#)
- [HPSC Outbreaks Hub](#) (high-level weekly updates)
- [IMSRL Annual Report](#)





IMD Reporting at EU level

Reporting to ECDC (for surveillance and cluster analysis)

- via [EMERT platform](#) for WGS data and EpiPulse Cases for case-based epi data
 - Data linked through field in data submissions 'NationalRecordID'
 - ECDC perform cluster analyses using cgMLST with cluster cutoff criteria
 - IMD clusters notified via EpiPulse to MS
 - Analyses available via Molecular Typing Tool for countries involved
 - Also [annual surveillance](#) and [ad hoc](#) reports



Cluster definition

Core genome multi-locus sequence typing (cgMLST)

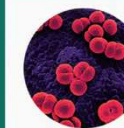
- Single-linkage cluster of more than two sequences within 10 allele difference (AD)
- For ST11, single-linkage cluster of more than two sequences within 5 AD
- Invasive isolates and EU/EEA countries

Fine typing

- Identical results from 10-loci fine typing scheme
- Invasive isolates and EU/EEA countries



Genomics and epi for extended IMD outbreak investigation



IMSRL
Irish Meningitis & Sepsis Reference Laboratory

SURVEILLANCE AND OUTBREAK REPORTS

A prolonged outbreak of invasive meningococcal disease in an extended Irish Traveller family across three Health Service Executive (HSE) areas in Ireland, 2010 to 2013

L O'Connor (lois_oconnor@hotmail.com)¹, M Ward¹, D Bennett², R Mulhall², P O'Lorcain³, R Cunney^{2,3}, R McDermott¹, E Neville⁴, J Heslin⁴, R FitzGerald⁵, K Meyler², M Conlon¹, A Clarke¹, B Corcoran⁶, G Fitzpatrick¹, B O'Connor⁴, P Flanagan³, D O'Flanagan³, S Cotter³

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O'Connor L, Ward M, Bennett D, Mulhall R, O'Lorcain P, Cunney R, McDermott R, Neville E, Heslin J, FitzGerald R, Meyler K, Conlon M, Clarke A, Corcoran B, Fitzpatrick G, O'Connor B, Flanagan P, O'Flanagan D, Cotter S. A prolonged outbreak of invasive meningococcal disease in an extended Irish Traveller family across three Health Service Executive (HSE) areas in Ireland, 2010 to 2013. Euro Surveill. 2015;20(21):pii=21139. Available online: <http://www.eurosurveillance.org/ViewArticle.aspx?ArticleId=21139>

Article submitted on 09 July 2014 / published on 28 May 2015

[O'Connor et al, 2015](#)



Journal of
Clinical Microbiology



Resolution of a Protracted Serogroup B Meningococcal Outbreak with Whole-Genome Sequencing Shows Interspecies Genetic Transfer

Robert M. Mulhall,^a Carina Brehony,^{b*} Lois O'Connor,^c Kenneth Meyler,^a Keith A. Jolley,^b James Bray,^b Desiree Bennett,^a Martin C. J. Malden,^b Robert Cunney^a

Irish Meningitis and Sepsis Reference Laboratory, Temple Street Children's University Hospital, Dublin, Republic of Ireland^a; Department of Zoology, University of Oxford, Oxford, England, United Kingdom^b; Department of Public Health, HSE East, Dr Steevens' Hospital, Dublin, Republic of Ireland^c

A carriage study was undertaken ($n = 112$) to ascertain the prevalence of *Neisseria* spp. following the eighth case of invasive meningococcal disease in young children (5 to 46 months) and members of a large extended indigenous ethnic minority Traveller family ($n = 123$), typically associated with high-occupancy living conditions. Nested multilocus sequence typing (MLST) was employed for case specimen extracts. Isolates were genome sequenced and then were assembled *de novo* and deposited into the Bacterial Isolate Genome Sequencing Database (BIGSdb). This facilitated an expanded MLST approach utilizing large numbers of loci for isolate characterization and discrimination. A rare sequence type, ST-6697, predominated in disease specimens and isolates that were carried ($n = 8/14$), persisting for at least 44 months, likely driven by the high population density of houses ($n = 67/112$) and trailers ($n = 45/112$). Carriage for *Neisseria meningitidis* ($P < 0.05$) and *Neisseria lactamica* ($P < 0.002$) (2-sided Fisher's exact test) was more likely in the smaller, more densely populated trailers. Meningococcal carriage was highest in 24- to 39-year-olds (45%, $n = 9/20$). Evidence of horizontal gene transfer (HGT) was observed in four individuals cocolonized by *Neisseria lactamica* and *Neisseria meningitidis*. One HGT event resulted in the acquisition of 26 consecutive *N. lactamica* alleles. This study demonstrates how housing density can drive meningococcal transmission and carriage, which likely facilitated the persistence of ST-6697 and prolonged the outbreak. Whole-genome MLST effectively distinguished between highly similar outbreak strain isolates, including those isolated from person-to-person transmission, and also highlighted how a few HGT events can distort the true phylogenetic relationship between highly similar clonal isolates.

[Mulhall et al, 2015](#)

The Outbreak:

Eight IMD cases from
provinces of Leinster
and Munster in
Republic of Ireland

3.5 year period

All cases were
children from the Irish
Traveller community

Aged 5 months to 3.5
years



April 2013 outbreak control team convened and following was instigated:

(a) **Alert** sent to all departments of public health

(b) **Further characterisation** by MLST, antigens PorA (MenB vaccine component) and FetA

(c) Communication with the manufacturers of a **Bexsero vaccine**

(d) Immunological testing of cases: **No immune deficiency** was identified



Co. Kildare
Case A: Mar 2010
Case B: Nov 2010
Case C: Mar 2011
Case E: Mar 2013
Case F: April 2013
Case H: Nov 2013

Co. Tipperary
Case D: Jan 2012
Case G: Jun 2013



Non-culture diagnosis and single gene typing



Case	Received Date	Age	County	<i>abcZ</i>	<i>adk</i>	<i>aroE</i>	<i>fumC</i>	<i>gdh</i>	<i>pdhC</i>	<i>pgm</i>	ST	clonal complex	PorA	FetA
A	09/03/2010	6 Months	Kildare	3	6	-	24	11	6	-	-	41/44	P1.7-2 4	F1-21
B	22/11/2010	5 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F1-21
C	10/03/2011	5 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F5-12
D	06/01/2012	10 months	Tipperary	4	5	2	-	-	-	20	-	-	-	-
E	21/03/2013	15 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F5-12
F	30/04/2013	3 Years	Kildare	-	6	9	24	-	6	-	-	41/44	P1.7-2 4	-
G	13/06/2013	9 Months	Tipperary	-	-	-	-	-	-	-	-	-	-	-
H	28/11/2013	6 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F5-12

Cases **B** and **H** yielded an isolate

Cases A, C, D, E, F and G were CSF or blood extracts



Partial profiles still provide information



IMSRL
Irish Meningitis & Sepsis Reference Laboratory

Case	Received Date	Age	County	<i>abcZ</i>	<i>adk</i>	<i>aroE</i>	<i>fumC</i>	<i>gdh</i>	<i>pdhC</i>	<i>pgm</i>	ST	clonal complex	PorA	FetA
A	09/03/2010	6 Months	Kildare	3	6	-	24	11	6	-	-	41/44	P1.7-2 4	F1-21
B	22/11/2010	5 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F1-21
C	10/03/2011	5 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F5-12
D	06/01/2012	10 months	Tipperary	4	5	2	-	-	-	20	-	-	-	-
E	21/03/2013	15 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F5-12
F	30/04/2013	3 Years	Kildare	-	6	9	24	-	6	-	-	41/44	P1.7-2 4	-
G	13/06/2013	9 Months	Tipperary	-	-	-	-	-	-	-	-	-	-	-
H	28/11/2013	6 Months	Kildare	3	6	9	24	11	6	9	6697	41/44	P1.7-2 4	F5-12

Cases **B** and **H** yielded an isolate

Cases A, C, D, E, F and G were CSF or blood extracts



Partial profiles still provide information

PubMLST

Public databases for molecular typing and microbial genome diversity

Case	<i>abcZ</i>	<i>adk</i>	<i>aroE</i>	<i>fumC</i>	<i>gdh</i>	<i>pdhC</i>	<i>pgm</i>	ST	clonal complex	PorA	FetA
D	4	5	2	-	-	-	20	-	-	-	-



Please select the scheme you would like to query:

MLST

Please enter your allelic profile below. Blank loci will be ignored. Autofill

<i>abcZ</i>	<i>adk</i>	<i>aroE</i>	<i>fumC</i>	<i>gdh</i>	<i>pdhC</i>	<i>pgm</i>
4	5	2				20

Options

Search:

Display/sort options

Order by:

Display: records per page

152 records returned (1 - 25 displayed). Click the hyperlinks for detailed information.



ST	<i>abcZ</i>	<i>adk</i>	<i>aroE</i>	<i>fumC</i>	<i>gdh</i>	<i>pdhC</i>	<i>pgm</i>	clonal complex
12	4	3	2	16	8	11	20	
56	4	10	2	8	8	11	20	ST-269 complex
96	4	10	2	16	8	11	20	ST-269 complex
336	4	5	2	9	21	11	20	
492	4	10	2	17	8	11	20	ST-269 complex
793	4	6	2	26	8	11	20	
794	4	6	2	17	8	11	20	
795	4	6	2	26	15	11	20	
797	4	10	2	9	8	124	20	ST-269 complex
798	4	10	2	55	8	11	20	ST-269 complex
799	4	10	2	17	3	11	20	
817	4	16	2	17	8	11	20	
963	57	5	2	35	6	68	20	
1342	4	10	2	135	8	11	20	ST-269 complex
1442	4	7	2	26	11	24	20	
1462	1	5	2	9	21	11	20	
1477	4	97	2	9	21	11	20	
1777	4	10	2	154	8	11	20	ST-269 complex
1879	4	5	2	5	38	11	1	
2086	4	5	2	16	8	11	20	
2210	4	6	2	196	8	11	20	
2307	4	5	2	5	38	11	9	
2369	4	10	2	161	8	11	20	ST-269 complex
2461	57	5	2	221	6	68	20	
2525	4	10	2	17	16	6	20	

<https://pubmlst.org/>



Public health response

Following case H in Nov 2013, 123 extended family members invited
112 (91%) attended clinic Dec 2013



Chemoprophylaxis offered to
all extended family



New (at the time) MenB
Bexsero® vaccine offered to
2 mths to 24 yrs



112 pharyngeal
swabs taken





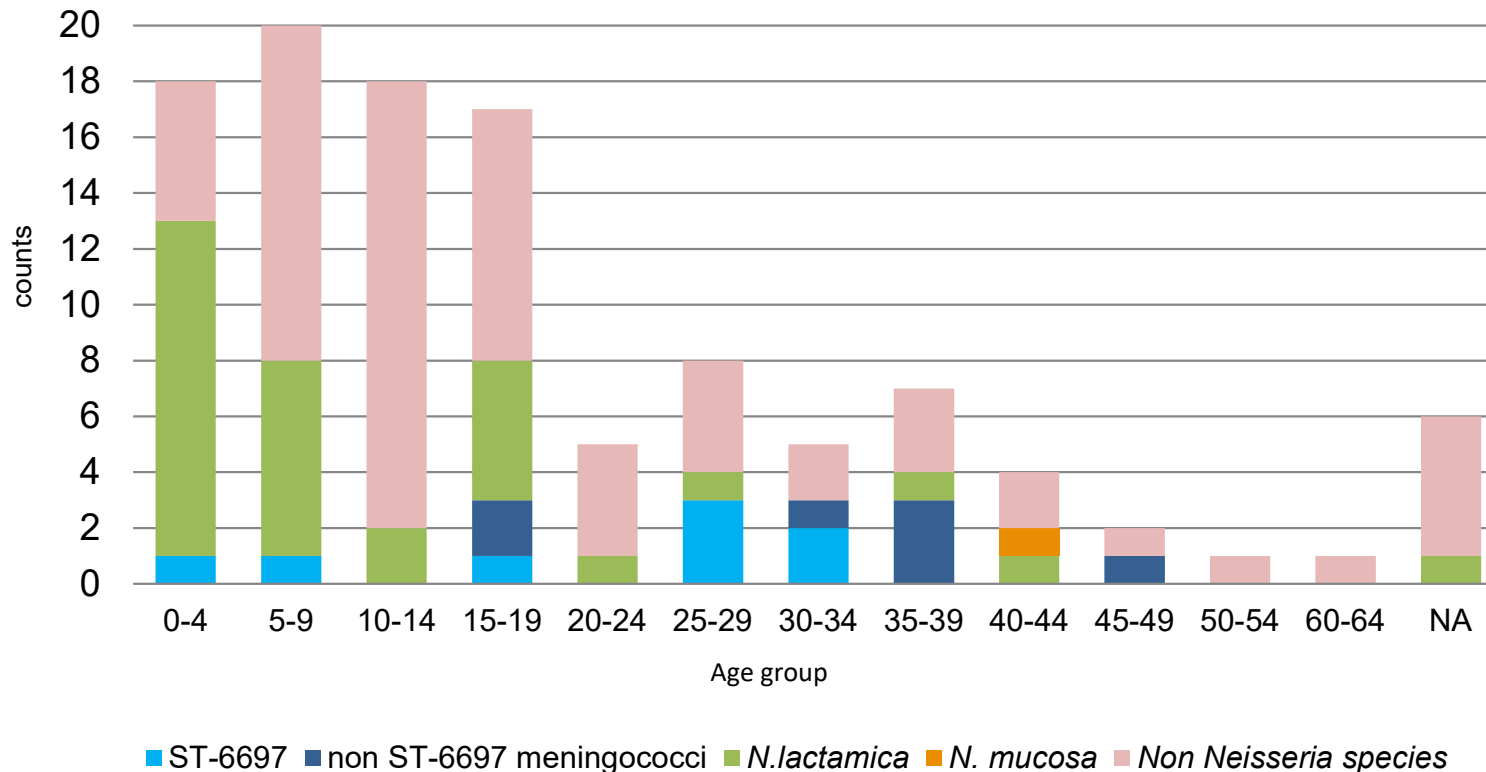
ST6697 strain also predominant in Nm carriers



ID	Age (Yrs)	Gender	HousingType	Serogroup	ST	Clonal complex	PorA	FetA
M1.1	28	Female	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	F5-12
M2.1	34	Male	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	F5-12
M13.1	16	Male	House	B	6697	ST-41/44cc	7-2,4	F5-12
M14.1	35	Female	Trailer/Caravan	B	7244	ST-461cc	19-2,13-1	F3-9
M16.1	15	Male	House	B	10659		22,9	F1-7
M20.1	38	Male	House	B	1163	ST-269cc	22,9	F1-7
M30.1	18	Male	Trailer/Caravan	W	1281	ST-22cc	18-1,3	F1-2
M31.1	32	Female	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	F5-12
M32.1	32	Male	Trailer/Caravan	B	10659		22,9	F1-7
M46.3	3	Male	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	F5-43
M45.1	6	Male	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	?
M34.1	27	Male	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	F5-12
M35.1	25	Female	Trailer/Caravan	B	6697	ST-41/44cc	7-2,4	F5-12
M42.1	38	Male	House	NG	1157	ST-1157cc	21-7,16	F5-36
PCR Positive Only	45	Male	House	B	-	-		-



Culture results of 112 swabs volunteered by extended Traveller Family: *Neisseria* spp. carriage



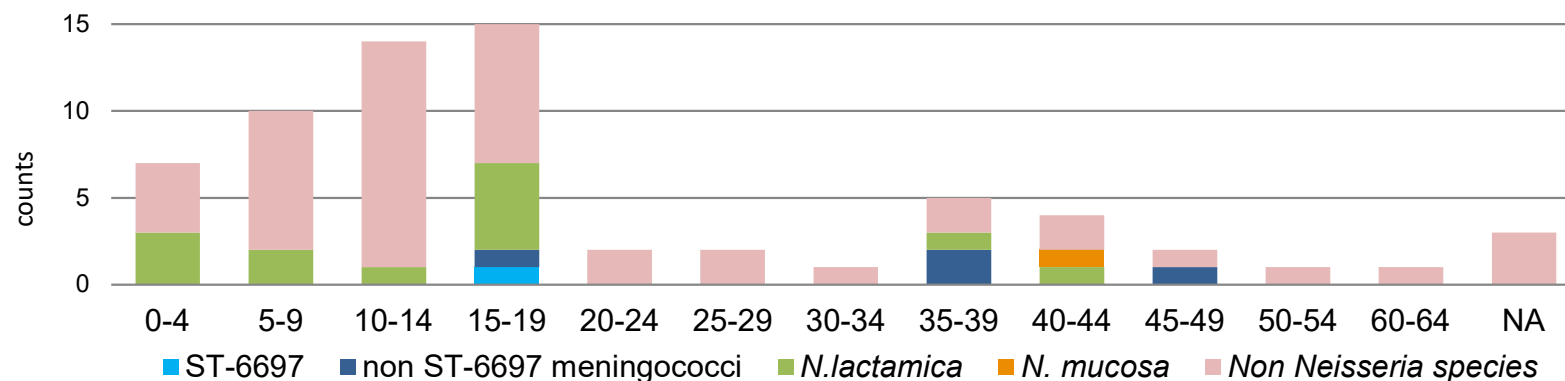
- Young age profile overall
- Commensal *N. lactamica* predominant in younger ages
- Nm predominant in older age groups
- Outbreak strain <9 yrs and 25-35



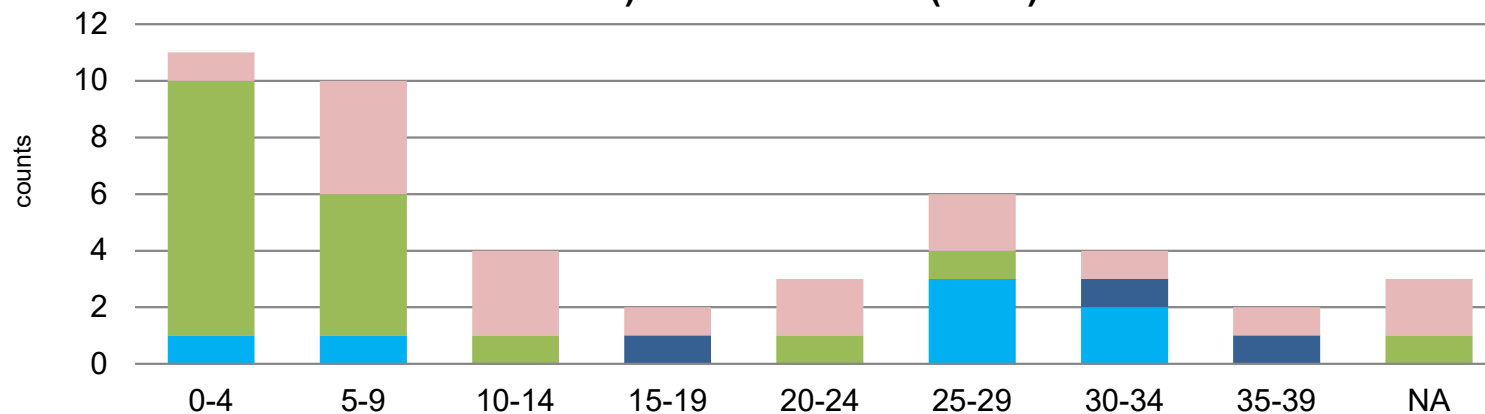
Impact of housing type on carriage of *Neisseria* spp.

67% of Nm carriage positive individuals lived in households of 8 or more

A) House (n=67)



B) Caravan/Trailer (n=45)



Whole Genome Sequencing of 2 case and 8 carried isolates



Jolley KA, Maiden MC. BIGSdb: Scalable analysis of bacterial genome variation at the population level.

[BMC Bioinformatics 2010, 11:595](#)

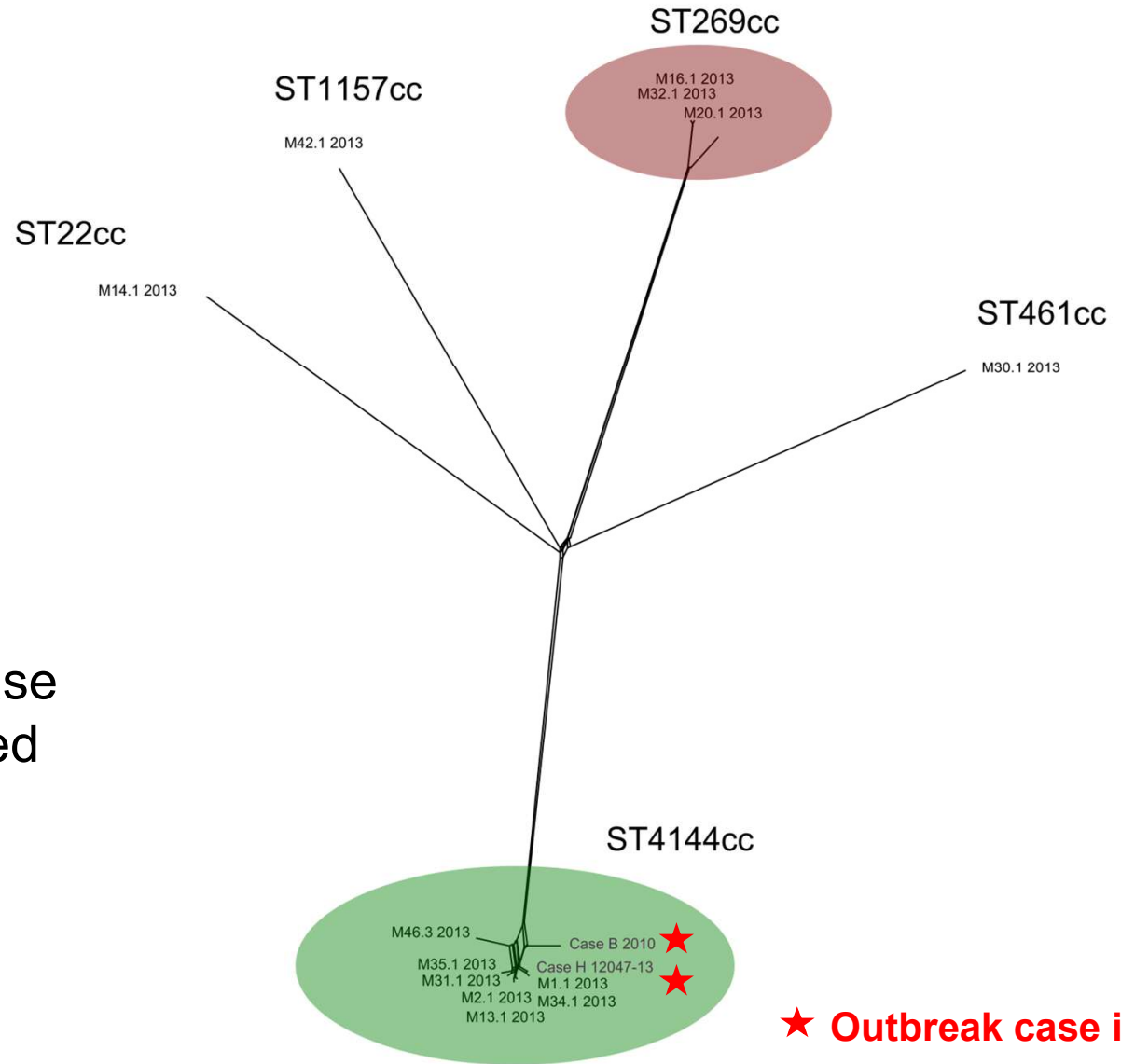
Bratcher HB, Corton C, Jolley KA, Parkhill J, Maiden MC. A gene-by-gene population genomics platform: de novo assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes.

[BMC Genomics 2014, 15:1138](#)



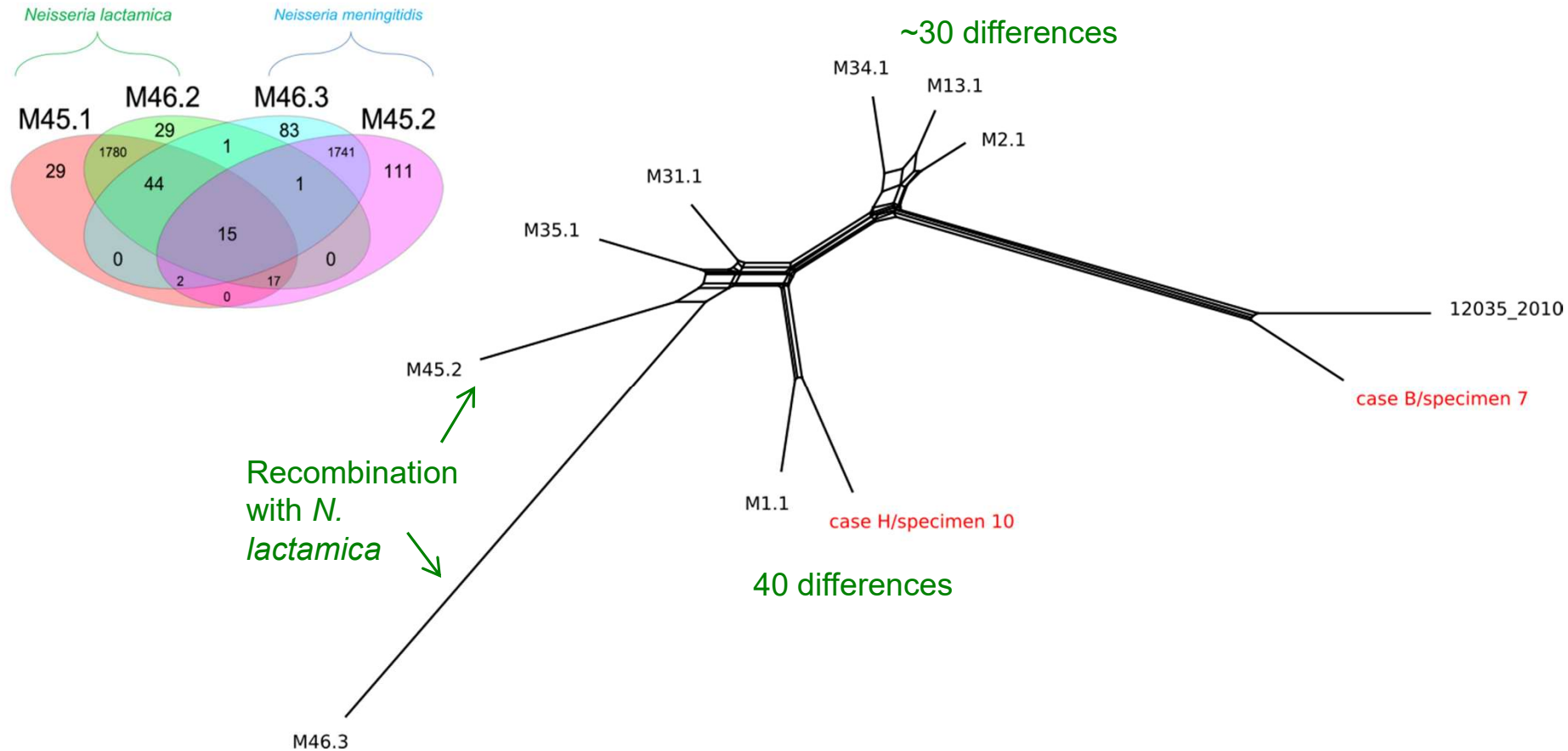
Core genome MLST
NeighbourNet diagram of case
and carrier isolates generated
from the **1605** core
meningococcal genes

100.0





Whole genome MLST NeighbourNet diagram of case and carried ST6697 isolates generated from the 2063 meningococcal genes





Summary: non-culture dx + typing and genomics for fine detail analysis

- Combination of non-culture diagnosis, culture, typing and WGS to resolve outbreak
 - Indicated uncommon serogroup B strain circulating in Traveller community for at least 3 years
 - Several risk factors involved especially housing
- Web-based tools (PubMLST) fast and ready to use
- Enhanced power of WGS added further insight and resolution into:
 - Outbreak detection, (re)emerging strains, evolution of strains
 - Interspecies recombination
 - Transmission dynamics
- PH, NRL and HPSC close collaboration





Clostridium difficile Infection (CDI) enhanced surveillance

- CDI case definition: clinical criteria + lab detection of toxin TcdA/B
- Cases notified to CIDR
- CDI data biannual/quarterly reporting from 65 hospital labs to HPSC
- *C. difficile* reference laboratory (Cherry Orchard, Dublin) (CDI NRL) sequence all CDI toxin +ve isolates
 - Send line listings to PH Depts and user labs fortnightly. Also, hospital specific reports e.g. re clusters as needed
- CDI NRL submit list of typed samples 2x per year to HPSC
 - Contains Sequence Type (ST), details of toxin genes and Cluster IDs
- List matched to CIDR notifications @HPSC. (CIDR doesn't have field for CDI genotype). Matched list shared with CDI NRL for analyses



<https://www.hpsc.ie/a-z/microbiologyantimicrobialresistance/clostridioidesdifficile/enhancedsurveillance/>



Clostridium difficile Infection (CDI) enhanced surveillance lab template form

Data from CDI NRL

Patient Details							Isolate Details			Onset Details			Origin Details			Severity		Typing		Treatment		
													Origin	Complete only if Healthcare associated	Complete only for Healthcare Exposure cases							
Hospital Code	Patient ID (required field - ID)	Age (if discharged, date of death, date of birth and date of admission)	Sex	Date of Birth	Patient admitted to hospital?	Date of admission	Case Type	Specimen ID (required field - ID)	Date Specimen Taken	Origin of Specimen	Onset location	Date of onset	Type of healthcare facility where onset occurred?	Origin	Type of healthcare facility where infection is associated?	Type of Ambulatory Treatment Only, complete for HE exposure: Ambulatory care cases (If discharged 4-12 weeks from HCF, do not complete)	Date of Ambulatory treatment commenced	ICU Admission	Surgery/Colectomy	Ribotype (only complete data, no time-phased)	Sequence type (e.g. ST1)	Treatment type



<https://www.hpsc.ie/a-z/microbiologyantimicrobialresistance/clostridioidesdifficile/enhancedsurveillance/>



CDI national and international reporting

- National Reporting
 - Notifications to CIDR
 - [HPSC Notifiable Disease Hub](#) (high-level weekly updates)
 - [HPSC Outbreaks Hub](#) (high-level weekly updates)
 - [CDI Biannual and Annual Reports](#) (Collaboration with Antimicrobial Resistance & Infection Control (AMRIC))
 - CDI NRL National Report
- Reporting to ECDC via EpiPulse platform
 - [ECDC Surveillance Atlas](#)
 - HPSC report yearly/2x per year upload of Irish data includes typing data i.e. ST, toxin gene
- EpiPulse Events/alerts monitored and information shared with CDI NRL e.g. fasta files/ENA accession





CDI surveillance 2025, Annual Report

- Matched CDI cases: 790 CDI cases matched between NRL and HPSC
- ST11, ST2, and ST8 most common sequence types
- Most strains were *tcdA* positive (93%) and *tcdB* (99%) positive
- 20% identified as being part of clusters, which were predominantly ST11 and ST8
 - 20 outbreaks: 16 with STs and 4 without predominant ST
- Slight differences amongst STs and healthcare association: ST8 (80%), ST2 (70%), ST11 (70%)
- Hypervirulent international clone ST1 present at v low prevalence (n=3)





CDI surveillance 2025, Annual Report

	Total cases		ST8		ST2		ST11	
	n	%	n	%	n	%	n	%
Total reported cases with sequence typing	790	-	98	12%	91	12%	73	9%
CDI toxin genotype								
<i>tcdA</i> positive	731	93%	98	100%	91	100%	31	42%
<i>tcdB</i> positive*	785	99%	98	100%	91	100%	71	97%
<i>cdtA/cdtB</i> positive	146	18%	0	0%	7	8%	73	100%
CDI cases identified as part of clusters	159	20%	24	24%	19	21%	17	23%
CDI Case Type								
– New	752	95%	96	98%	86	95%	71	97%
– Recurrent	34	4%	2	2%	4	4%	2	3%
– Unknown	4	1%	0	0%	1	1%	0	0%
CDI Origin								
– Healthcare-associated (HCA)	588	74%	78	80%	64	70%	51	70%
– Community associated (CA)	161	20%	14	14%	21	23%	17	23%
– Discharged 4-12 weeks from HCF	37	5%	5	5%	6	7%	5	7%
– Unknown	4	1%	1	1%	0	0%	0	0%
CDI Severity								
Critical care admission or colectomy	19	2%	2	2%	4	4%	1	1%

<https://www.hpsc.ie/a-z/microbiologyantimicrobialresistance/clostridioidesdifficile/cdifficiledataandreports/annualreports/>



CDI NRL Annual Report, 2025

- CDI NRL (Cherry Orchard Hospital, Dublin) operating since Q2 2022
- N=1446 PCR toxin +ve CDI samples sequenced in 2025
- Toxin gene detection (and other putative virulence factors)
 - *tcdA* detected in 88.7% and *tcdB* in 98.9%
- AMR determinant detection
 - *tet(M)* detected in 15.3%, *ermB* 7.1%
 - Genotypic AMR gene detection replaced phenotypic AST 2024
- Sequence Type (ST) assignment
 - Clonal complex 1 (CC1) 81.8% isolates
- cgMLST cluster analysis*
 - Cluster reported to PH ≤ 2 ADs
 - 216 clusters in 2025

*Bioinformatic analyses performed using Ridom SeqSphere+ platform



Figure 13. Minimum spanning tree (MST) of 2025 Isolates submitted to the Reference Laboratory Service. The tree is colour coded by sequence type (ST) and illustrates how each ST is grouped together.



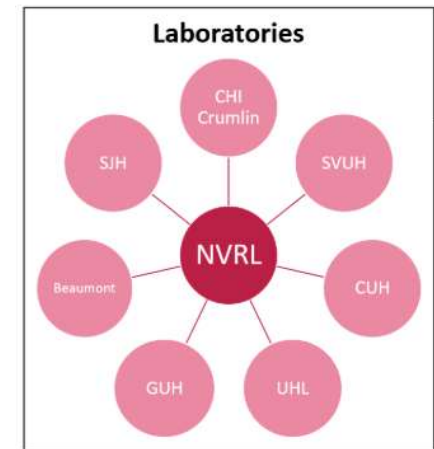
Other epi and typing/genomic collaborative surveillance

- VTEC, *Campylobacter*, *Salmonella*, *Shigella*, *Listeria*, CPE, *N. gonorrhoeae*, *S. pneumoniae*...
- Viruses: Influenza, SARS-CoV-2, RSV, Mpox, HIV, measles, norovirus...
- Wastewater surveillance: SARS-CoV-2 viral load, SARS-CoV-2 variants, influenza, RSV, polio, AMR (planned)



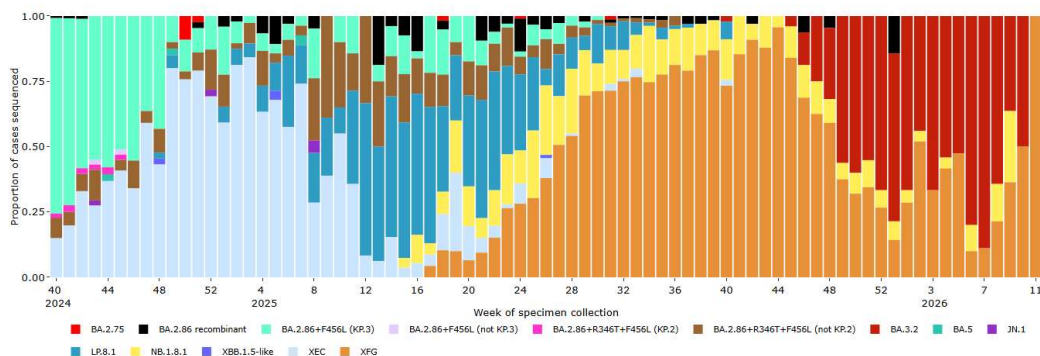
First national genomic surveillance Programme: SARS-CoV-2 whole genome sequencing, Ireland

- National SARS-CoV-2 WGS programme officially closed in 2025 – reverted to BAU WGS Programme – HPSC
- Hub & spoke model – the NVRL (hub) & 6/7 regional hospital labs (spokes)
- Previous programme successes – global metrics – TAT, % cases sequence, GISAID, ECDC reporting
 - ~120,000 SARS-CoV-2 genomes sequenced
- Sampling framework revised – frequency and volume
- Improvements to genomic surveillance data linkage and reporting
- With reduced numbers, reporting will be revised
- Integrate with influenza and RSV
- 2026/27 – integrated resp. virus sampling framework (NVRL only)

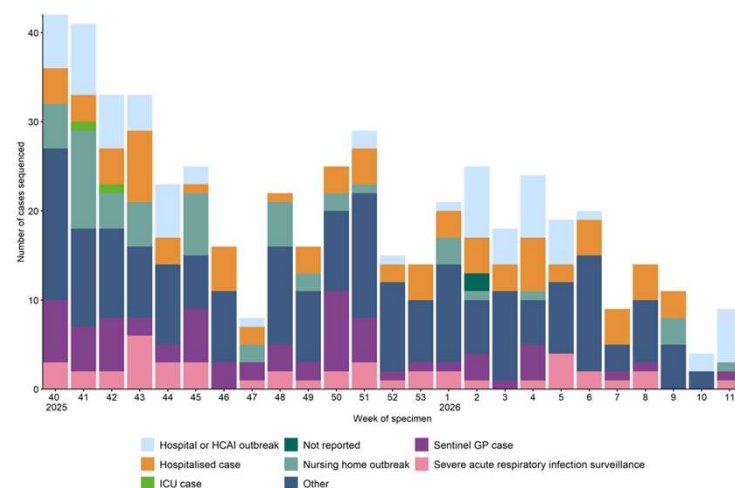
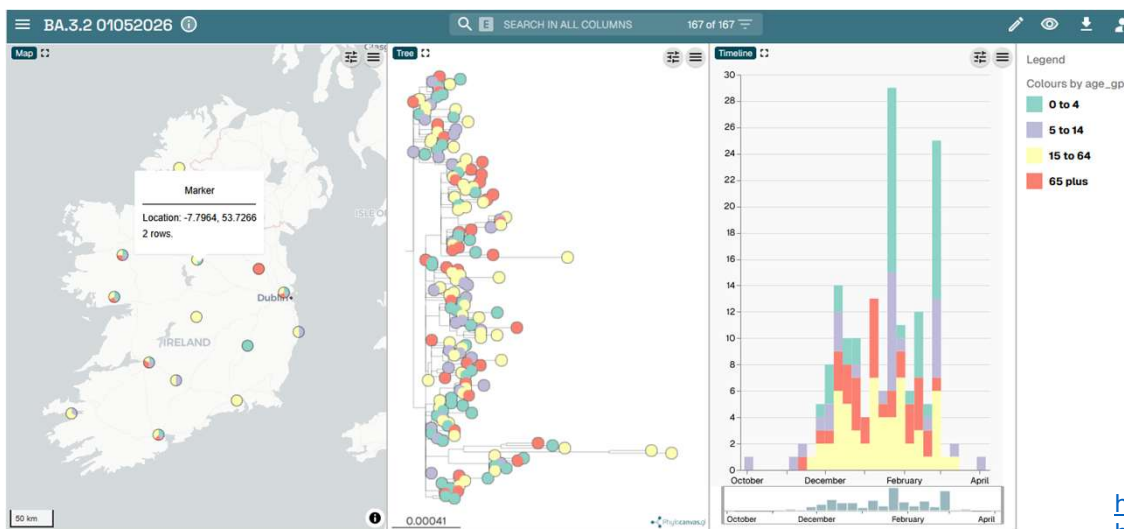




First national genomic surveillance Programme: SARS-CoV-2 whole genome sequencing, Ireland



Country / territory	Sequences shared	Reported COVID-19 cases	Reported deaths from COVID-19	% of cases sequenced and shared
Canada	34,927	5,662	89	616.867
Denmark	3,875	8,276	242	46.822
Sweden	5,828	14,405	789	40.458
France	9,447	28,926	0	32.659
China	4,981	22,186	150	22.451
Ireland	3,099	13,946	251	22.221
Luxembourg	1,007	4,755	0	21.178
Mexico	1,462	6,992	323	20.91
New Zealand	2,349	56,014	434	4.194
Portugal	1,029	24,879	880	4.136
Italy	3,881	102,070	464	3.802
Greece	1,341	97,757	659	1.372
Russia	2,175	426,069	746	0.51
Israel	1,795	0	0	



<https://respiratoryvirus.hpsc.ie/>
<https://www.hpsc.ie/a-z/respiratory/acuterespiratoryinfection/surveillance/integratedreports/>



SARS-CoV-2 Epi and Genomic data application for European surveillance

Case-based epi data ECDC EpiPulse

NationalRecordId
DateUsedForStatistics
SubjectCode
HealthTopic
DataSource
ReportingCountry
Status
Sex
Age
AgeMonth
DateOfOnset
Hospitalisation
DateOfHospitalisation
DateOfDischarge
IntensiveCare
DateOfICUHDU
DrugUsedProphylaxis
DrugUsedTreatment
RespSupport
ComplicationDiagnosis
HealthCareWorker
Outcome
DateOfDeath
WgsAccession
DateOfSpecimen
DateOfCOVID19Vaccination
COVID19Vaccination
EHRDiagnosticCode
DateLastVaccDose
InfluenzaVaccination
Pathogen
Coinfection
InfluenzaTypeSubtype
CoinfectionOther
SurveillanceType
CasePrecondition

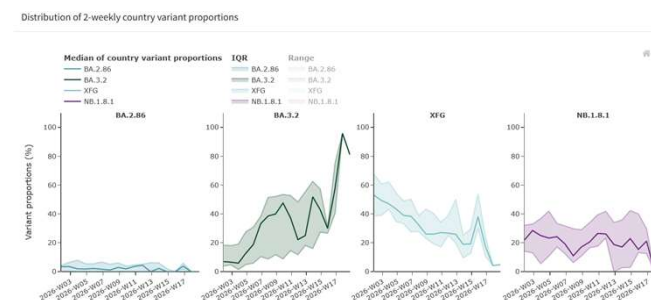


GISAID sequence data

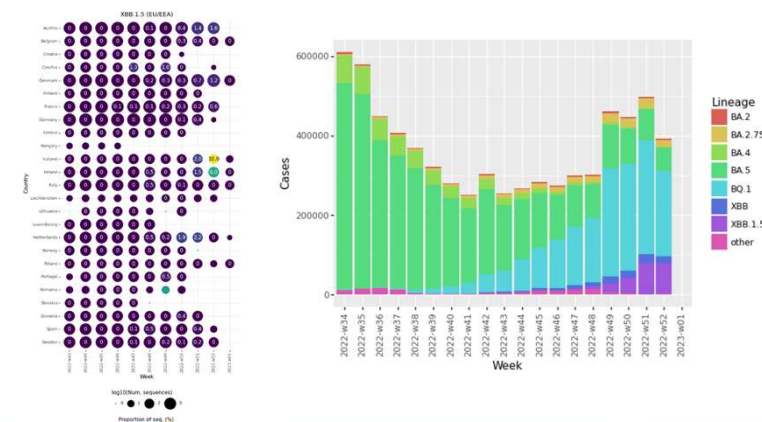
Registered Users	EpiFlu™	hCoV-19	EpiRSV™	EpiPox™	EpiArbo™	EpiInfl™	My Profile
EpiCoV™ Search Downloads Upload							
Search ▼ Reset filters							
EPI_ISL ID	Virus name	EPI_SET ID	Complete	High coverage	Low coverage excluded	With patient status	Collection date complete
Location	to	Host	Under investigation				
Collection	to	Submission					
Clade	all	Lineage	Variant				
AA Substitutions		Nucl Mutations					
Text Search							
☐	VIRUS NAME	LINEAGE	ACCESSION ID	COLLECTED	SUBMITTED	LEN	HOST
☐	hCoV-19/Ireland/D-BH-04282006/2026	RV.1.3	EPI_ISL_20478543	2026-01-06	2026-06-19	29,649	Human
☐	hCoV-19/Ireland/D-BH-04187149/2026	XFG.14.8	EPI_ISL_20478542	2026-05-29	2026-06-19	29,649	Human
☐	hCoV-19/Ireland/D-BH-04295091/2026	XFG.3.3	EPI_ISL_20478541	2026-04-01	2026-06-19	29,648	Human
☐	hCoV-19/Ireland/D-BH-02680087/2026	XFG.3.3	EPI_ISL_20478540	2026-03-28	2026-06-19	29,648	Human
☐	hCoV-19/Ireland/D-CS260032/2026	RV.1.3	EPI_ISL_20468261	2026-04-03	2026-06-15	29,819	Human
☐	hCoV-19/Ireland/D-CS260031/2026	SK.1	EPI_ISL_20468260	2026-03-29	2026-06-15	29,717	Human
☐	hCoV-19/Ireland/D-CS260030/2026	QF.2.1	EPI_ISL_20468259	2026-04-03	2026-06-15	29,836	Human
☐	hCoV-19/Ireland/D-CS260029/2026	XFG.1.1.3	EPI_ISL_20468258	2026-04-13	2026-06-15	29,832	Human
☐	hCoV-19/Ireland/D-CS260025/2026	XFG.1.1.3	EPI_ISL_20468257	2026-04-17	2026-06-15	29,834	Human
Total: 191,216 viruses							
Subsample Charts EPI_SET Select Analysis Download							



Surveillance: ECDC/WHO EURO ERVISS



Variant Analysis and Risk Assessment





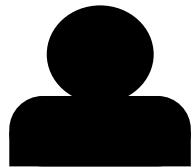
ECDC EpiPulse



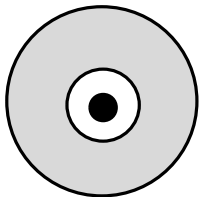
Data sharing with ECDC: challenges and opportunities

(slides shared with permission from ECDC)

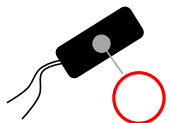
Which data do we collect?



Epidemiological data (e.g. age, gender, clinical presentation, travel history, ...)



Molecular data

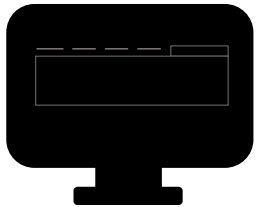


Phenotypic data (e.g. AMR data, serotype, ...)

Genotypic data (e.g. ST, ...)

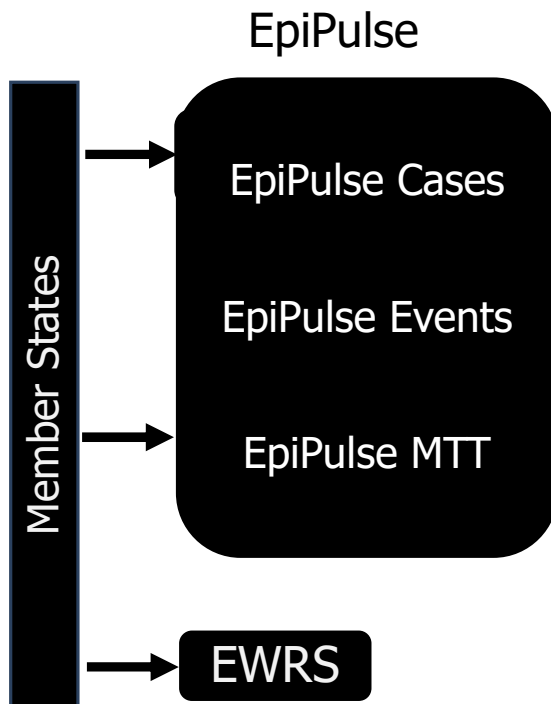
Genomic data (e.g. reads, assemblies, accessions, ...)

EpiPulse (the European surveillance portal for infectious diseases)



- EpiPulse is an application that allows interactions between public-health institutes and ECDC
- It consists of several modules (e.g. events, cases, molecular typing tool)
- Domains span all the infectious diseases covered by ECDC
- Access: nominated experts from Member States and selected countries outside the EU/EEA which have agreed cooperation frameworks with ECDC, ECDC staff and global partners (e.g. WHO EURO staff). Access rights vary depending on each person's assigned role(s)

How do we collect the data?



Designed for

Collecting, analysing and disseminating surveillance data [structured data; allows for uploading large batches of data; entry point for indicator-based surveillance data]

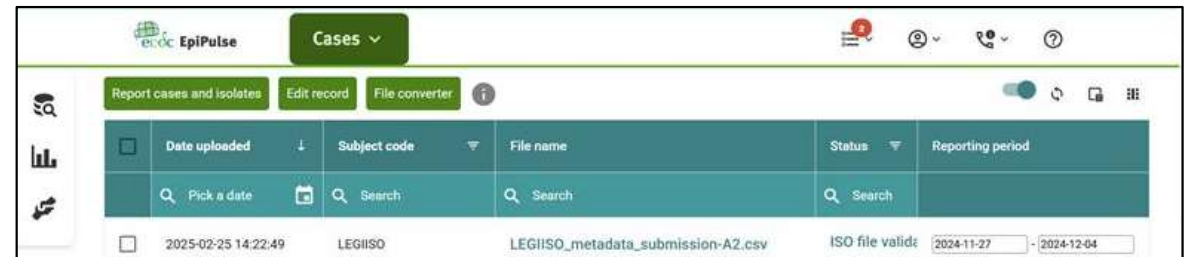
Collecting, analysing, sharing, and discussing infectious disease data for threat detection, monitoring, risk assessment and outbreak response. Data are shared on a voluntary basis [event-based surveillance]

Visualize WGS data

Formal notification of outbreaks, clusters and contact tracing data. Data are used for risk management

EpiPulse Cases

- A new system for data submission to ECDC, with better support for WGS data compared to TESSy
- Epidemiological data can be uploaded as csv or xml files
- Genomic data can be uploaded as assembly (fasta) or raw reads (fastq.gz, preferred option)
- The genomic data and epidemiological data are linked through the use of naming conventions that link the RecordId and the file names
- Files can be uploaded through a web portal or using an API

The screenshot shows the EpiPulse web portal. At the top, there's a header with the 'EpiPulse' logo and a 'Cases' dropdown menu. Below the header, there are three main buttons: 'Report cases and isolates', 'Edit record', and 'File converter'. A sidebar on the left contains icons for search, a bar chart, and a person. The main content area features a table with columns: 'Date uploaded', 'Subject code', 'File name', 'Status', and 'Reporting period'. Each column has a search icon and a search input field. Below the table, there's a row of data: a checkbox, the date '2025-02-25 14:22:49', the subject code 'LEGISO', the file name 'LEGISO_metadata_submission-A2.csv', the status 'ISO file valid', and the reporting period '2024-11-27 - 2024-12-04'.

EpiPulse [home page]



Events ▾

Molecular Typing



i Dear TESSy and EpiPulse Cases users,

As of March 31, 2025, the collection of EVD, FWD, and LEGI data has transitioned from TESSy to EpiPulse Cases.
For more information, please refer to the Help section.

Thank you!



Welcome to EpiPulse - The European surveillance portal for infectious diseases!

Tutorials on how to use EpiPulse



Get started



How to create an item



How to create a comment



Advanced search

Manage cookies

EpiPulse Events

**Events** ▾

Molecular Typing



Please treat the data in the platform as sensitive non-classified unless specifically indicated as public.

All ARHAI EI LEGI EVD FWD HEP HIV IRV PREP SoHO SRV STI **TB** VPD Resp. duty TALD Daily CDTR Weekly CDTR

Search by text

in Please select



Announcement

Event

Forum

Long-Term Monitoring

News

Signal

Threat

Closed

Discarded

Open

⊕ Advanced search criteria

Create item

Create output

Edit flags



Column visibility

Show 25 rows



	ID	Participating domain	Type	Title	Created by	Pathogens	Diseases	Modified time	Flags
☐	2025-TUB-00001	EI, TB	Event			Mycobacterium tuberculosis	Tuberculosis	2025-04-08 18:04	
☐	2025-TUB-00002	EI, TB	News	World Tuberculosis Day - 2025	ECDC/Public Health	Mycobacterium tuberculosis	Tuberculosis	2025-03-28 11:13	
☐	2024-TUB-00004	TB	Forum	Occupational TB screening	Hungary/Public Health	Not applicable	Not applicable	2024-12-12 13:35	

Showing 1 to 3 of 3 entries

Previous **1** Next

The information contained in this page is not public. In the ID column items are marked with the word (New) if they have been created in the last 7 days.

[Manage cookies](#)

EpiPulse Events



Events ▾

Molecular Typing



Please treat the data in the platform as sensitive non-classified unless specifically indicated as public.

View access settings



< Previous

Next >

Item details

ID: 2025-TUB-00001 Type: Event Title: [REDACTED]

Diseases: Tuberculosis Pathogens: Mycobacterium tuberculosis Participating domain: EI, TB

[Key information](#)
[Document workspace \(contains 0 files in 0 spaces\)](#)

Situation Awareness

Comments

Messages

Links

Outputs

Molecular Typing

Visualisations

ECDC

Add/Edit text box

Create/update comment

View history



Item created on: 2025-02-10 15:09

Item last updated on: 2025-04-08 18:04

Number of reply comments: 14

NCC: Number of confirmed cases: 47

NNC: Number of non-confirmed cases: 0

NCD: Number of confirmed deaths: 0

Show 10 rows

View comment

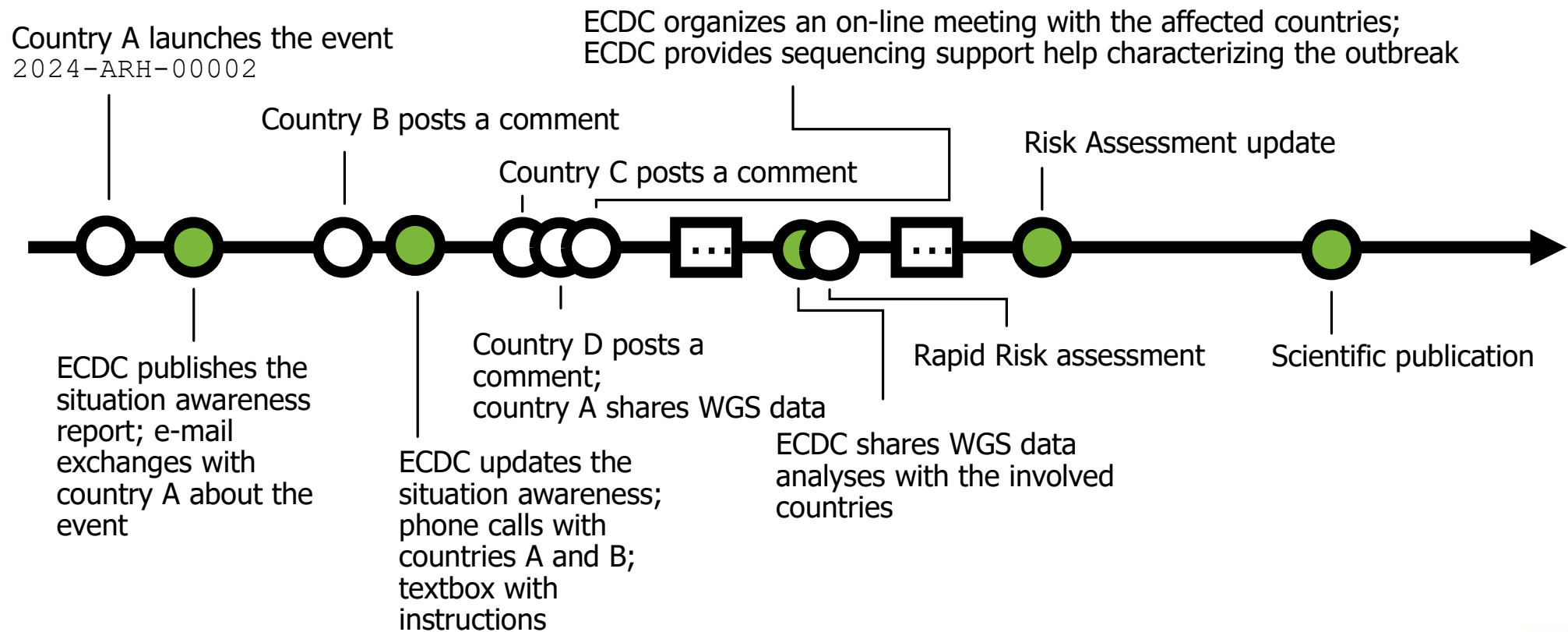
☐	Domain	Country / Organisation	Sector	NCC	NNC	NCD	Epidemiological information	Microbiological information	Additional information	Modified time	
☐	TB	[REDACTED]	Public Health	14	0		[REDACTED]			2025-02-14 11:01	

Manage cookies

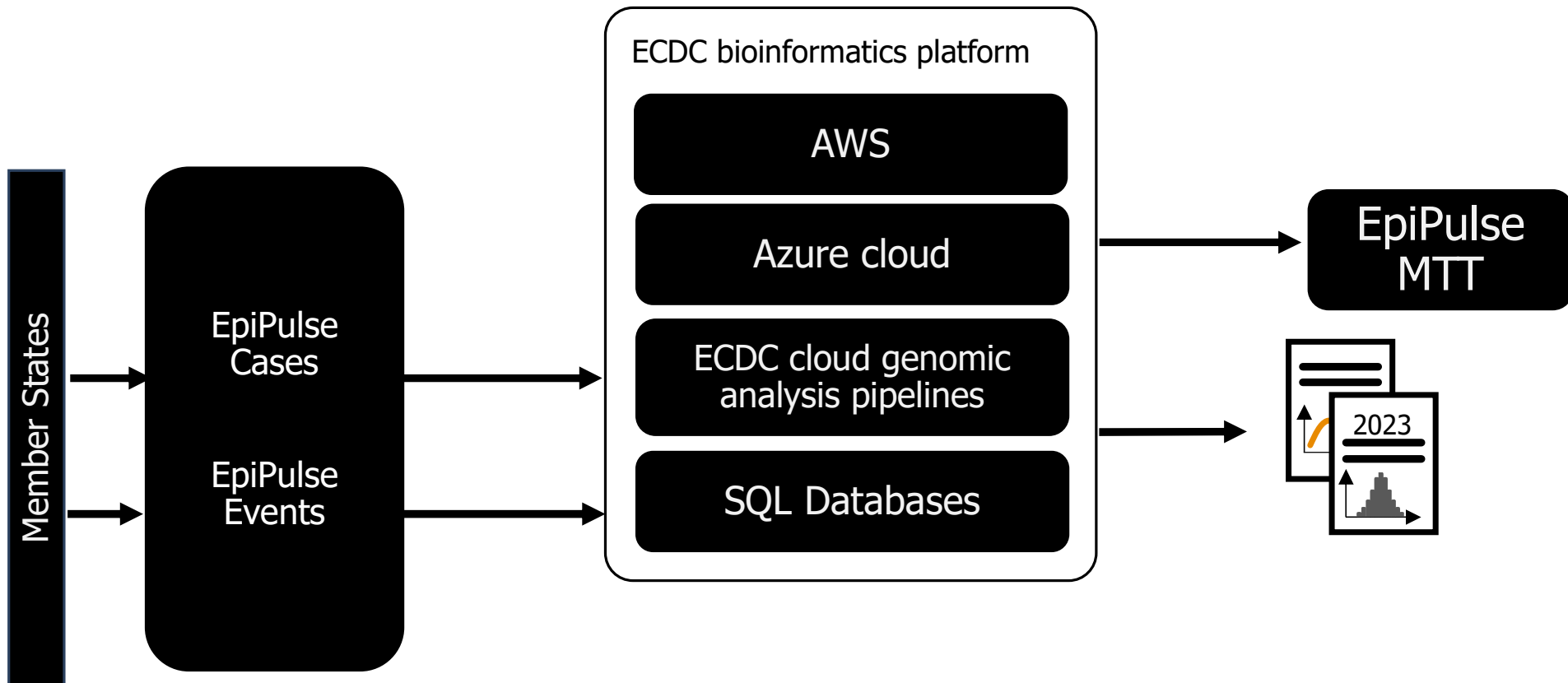
EpiPulse Events items

- **Signals:** cases/clusters/outbreaks/incidents which are assessed as not posing a public health risk for the EU/EEA (originate from epidemic intelligence media monitoring or automated analyses of indicator-based surveillance data, e.g. clusters detected by ECDC)
- **Events:** cases/clusters/outbreaks/incidents that are assessed to pose or may pose a public health risk for the EU/EEA (*Salmonella* cluster reported by country X)
- **Threats:** cases/clusters/outbreaks/incidents that are assessed to pose a public health risk for the EU/EEA and are of concern (e.g. new flu variant and high number of deaths)
- **Forum:** questions or requests for which replies from the other users are expected (e.g. question on the specificity of a laboratory method)
- **News:** communications or announcements (e.g. ECDC announcements, conferences and meetings)

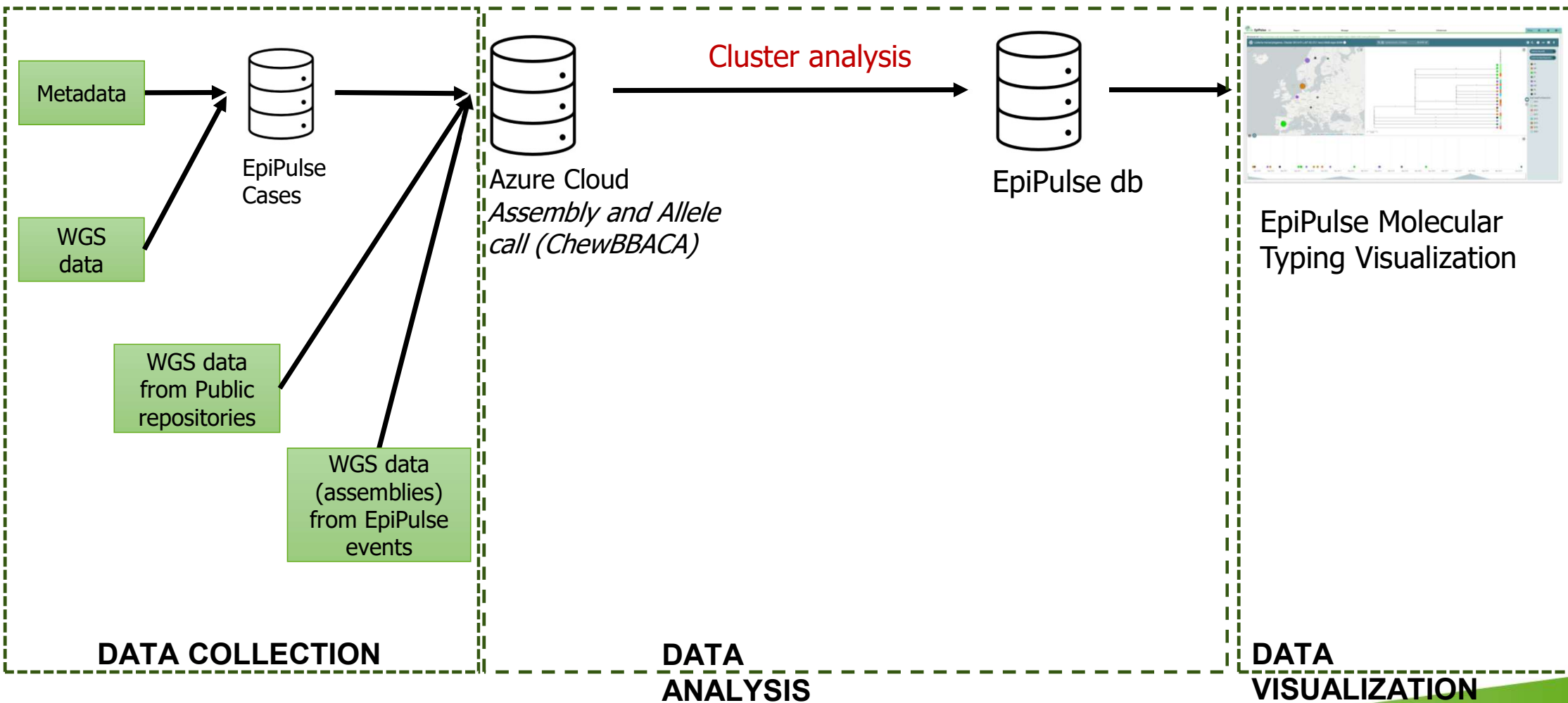
EpiPulse flow



ECDC data ecosystem



ECDC system

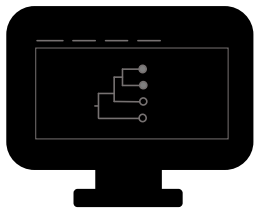


Analysis tools

Pathogen	schema	
<i>Listeria monocytogenes</i>	Chewie-NS	ResFinder, PlasmidFinder, MLST
<i>E.coli</i>	Chewie-NS	ResFinder, PlasmidFinder, MLST, VirulenceFinder
<i>Salmonella enterica</i>	Chewie-NS	Seqsero, ResFinder, PlasmidFinder, MLST
<i>Klebsiella</i>	Ridom	Kleborate, PlasmidFinder, MLST
<i>Mycobacterium tuberculosis</i>	Ridom	ResFinder
<i>Legionella pneumophila</i>	Ridom	ResFinder, El_gato (under implementation)
<i>Campylobacter</i>	PubMLST	MLST, ResFinder
<i>Clostridium difficile</i>	Ridom	MLST, ResFinder
<i>Providencia</i>	Custom	MLST, ResFinder, PlasmidFinder
<i>Acinetobacter baumannii</i>	PubMLST	MLST, ResFinder, PlasmidFinder
<i>Corynebacterium diphtheriae</i>	PubMLST	Custom
<i>Yersinia</i>	PubMLST	MLST
<i>Enterobacter hormaechei</i>	Ridom	MLST, ResFinder, PlasmidFinder

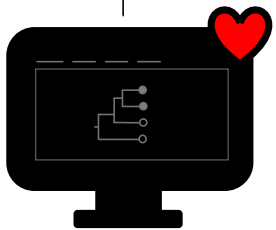
- Prefetch from NCBI
- Fastp
- quast
- Rasusa
- Spades
- ChewBBACA
- Allele QC
- Cluster analysis
- Visualization of clusters (Single-linkage, NJ)

EpiPulse: the Molecular Typing Tool (MTT)



- When genomic sequences are shared with ECDC, they are automatically analysed to:
 - find if they are genetically related to other known sequences (*i.e.* are part of a cluster) → can generate signals. Cut-offs to define clusters depend on the typing scheme used.
- The Molecular Typing tool is the EpiPulse module to visualize and analyse genetic clusters
- This is particularly relevant in the context of events / outbreaks → overall situation
- It is possible to associate clusters identified by the molecular typing tool to EpiPulse events

The Molecular Typing Tool... is for you!



The Molecular Typing tool allows you to explore and put in context your own isolates. For instance:

- You can get AMR and typing data. All this info can be exported.
 - TB: lineage calls, resistance predictions
- You can perform queries using your isolates ID and even set your own AD thresholds
- The tool is constantly being improved

The Molecular Typing Tool



Events ▾

Molecular Typing

i Dear TESSy and EpiPulse Cases users,

As of March 31, 2025, the collection of EVD, FWD, and LEGI data has transitioned from TESSy to EpiPulse Cases.
For more information, please refer to the Help section.

Thank you!



Welcome to EpiPulse - The European surveillance portal for infectious diseases!

 ▾

Tutorials on how to use EpiPulse



Get started



How to create an item



How to create a comment



Advanced search

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The Molecular Typing Tool: clusters



Events ▾

Molecular Typing



Please treat the data in the platform as sensitive non-classified unless specifically indicated as public.

Please, select a Pathogen to review and analyse its data.

Mycobacterium tuberculosis ▾

[➤ Search & Refine Selection](#)

Molecular Clusters

Visualisation

My Isolates Data

These table shows molecular clusters detected by ECDC by applying automated algorithms, see respective surveillance protocols for details. Only multi-country clusters & clusters in your country are shown. Included are also generated datasets of particular interest.

Mycobacterium tuberculosis clusters

Copy

Search

Column groups visibility ▾

Columns visibility ▾

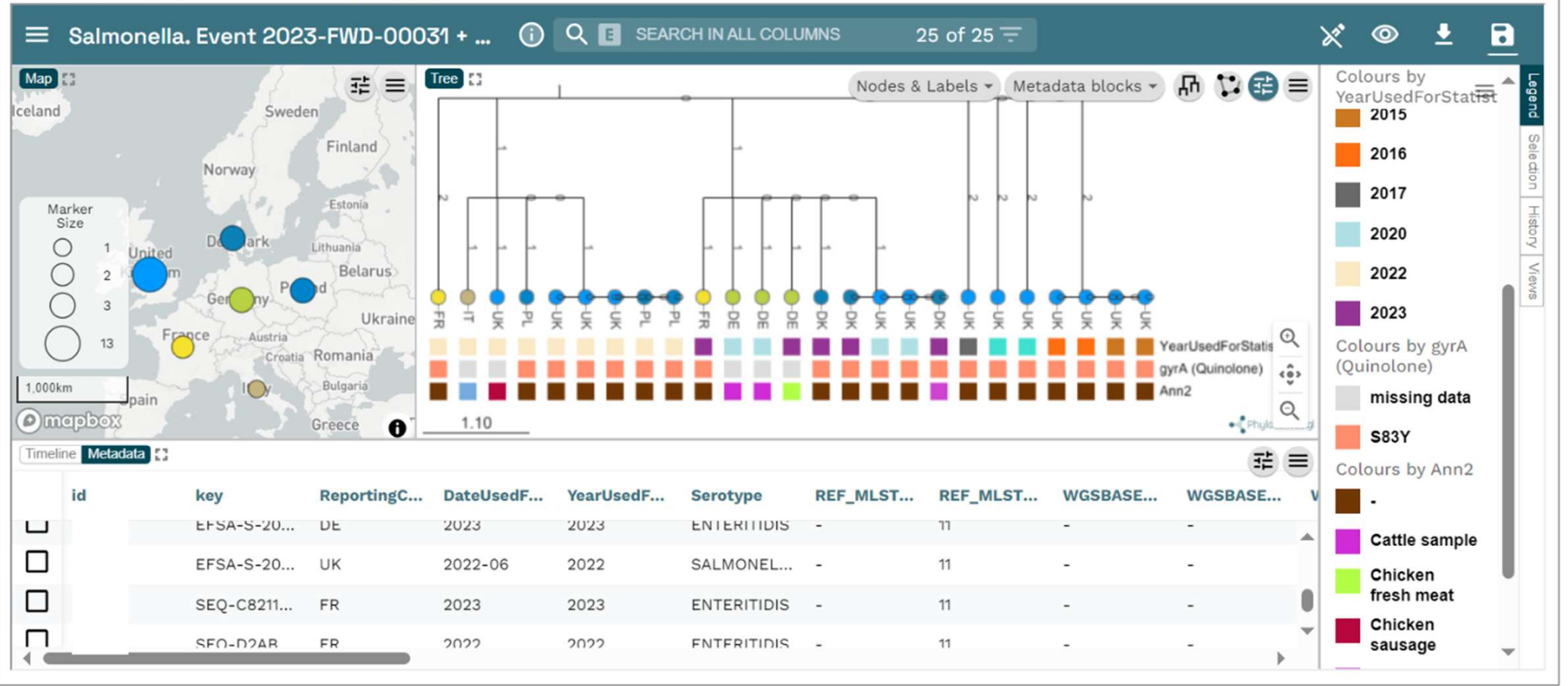


Clear filters

	Cluster ID	Clustering method	Cluster info	Event ID	Cluster status	Number of isolates	Countries	Updated since last analysis	First date
☐									
☐	2025-02.TB.03	cgMLST ChewBBACA			CLOSED	2	EE(2)	EE(2)	2024-06-14

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Microreact Url: <https://microreact.ecdc.europa.eu/project/ihborcHvksS3LzeFbbUrUD-salmonella-event-2023-fwd-00031-matches-within-5>



ECDC Learning Portal training resource

- Free Open Access ECDC Learning Portal
<https://learning.ecdc.europa.eu/>
- Live online/recorded sessions, other material
- In particular... GenEpi BioTrain series
 - 'Bacterial Strain Taxonomy for Genomic Surveillance'
 - 'Virulence Profiling of *E. coli* Pathotypes'
 - 'Genomic Epidemiology and Outbreak Investigation of Brucellosis in a One Health Context'



**GenEpi-BioTrain - Virtual Training 27 -
Clinical application of Nanopore-based 16S
metagenomic sequencing for diagnostics**

You can enrol in this course

[View course](#)

Acknowledgements

This training material was created by ECDC with the direct involvement of Erik Alm, Rodrigo Filipe, Luca Freschi, Andreas Hoefer, Priyanka Nannapaneni and Maximilian Riess.



National Infectious Disease Outbreak, Case, and Incident Management System (OCIMS)



What is OCIMS?

- Outbreak, Case, Incident Management & Surveillance System. Successor to CIDR
- Approved & Funded by Department of Health, following pandemic
- One **national centralised** public health system that will encompass All Hazards
- One **modern integrated system**, reducing silos, consolidating HSE information and reporting
- Improve **communication** between national and regional multidisciplinary teams both in and out of hours
- Business Case 2022 describes OCIMS as:

“This solution will support the needs of Public Health practitioners to identify, capture, respond, close and report on cases and outbreaks and will provide staff with electronic access to relevant and meaningful patient health information. The availability of this data will also enable staff to track and report on trends, enabling further public health activity”





Key Features of OCIMS

See previous limitations of current model. New model will:

- Move to integrated, case-based surveillance model
- Automate data validation and linkage
- Linking with external systems e.g. Lab, epi, and genomics data
 - Captured in a single system or seamlessly linked
 - Automatically ingested from source systems (labs, sequencing): HL7, APIs, NIS, Geocoders
 - Structured using standardised identifiers and data models

New model will enable:

- Near real-time integration and analysis
- Automated cluster detection and signal generation
- Improved, consistent reporting at national and international level
- Potential for GIS mapping
- Plus, other features which enhanced data collection, collation, action and reporting



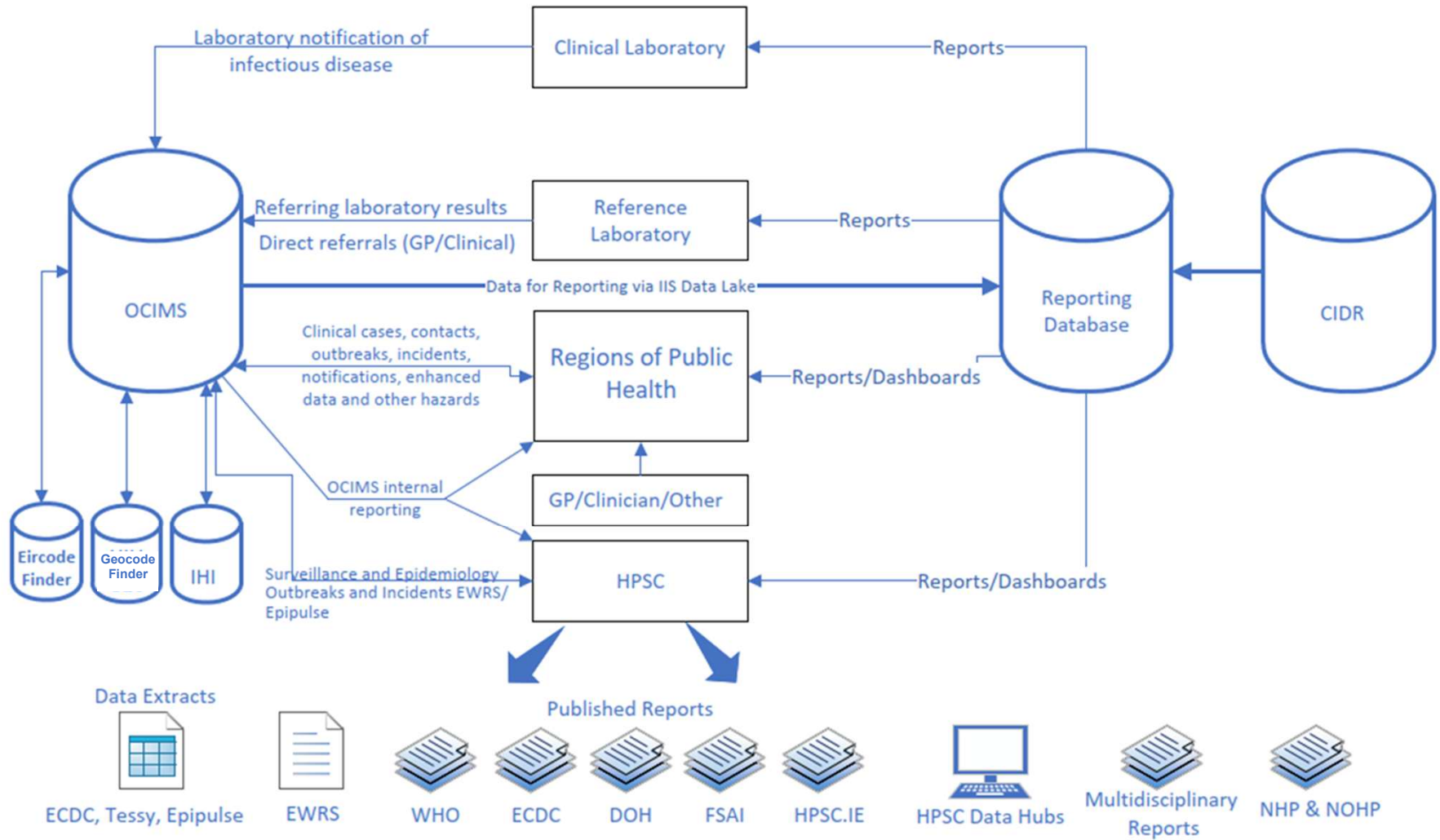


Summary: Current state of play and future goals

- Legacy systems at end of life (CIDR)
- Good links between national PH and Clin Micro and NR Labs
- Mix of automated and manual linking within CIDR and out
- Ongoing efforts to improve while also onboarding new systems
 - OCIMS dev.
 - ECDC EpiPulse etc
- National Integrated PH systems (OCIMS, NIIS...)
- Centralised NRL System/Campus, MedLIS
- Closer to real-time integration
 - Improved collaboration
 - Improved timeliness of information for action
- Strengthen for legacy use for surveillance and national/in'tl PH



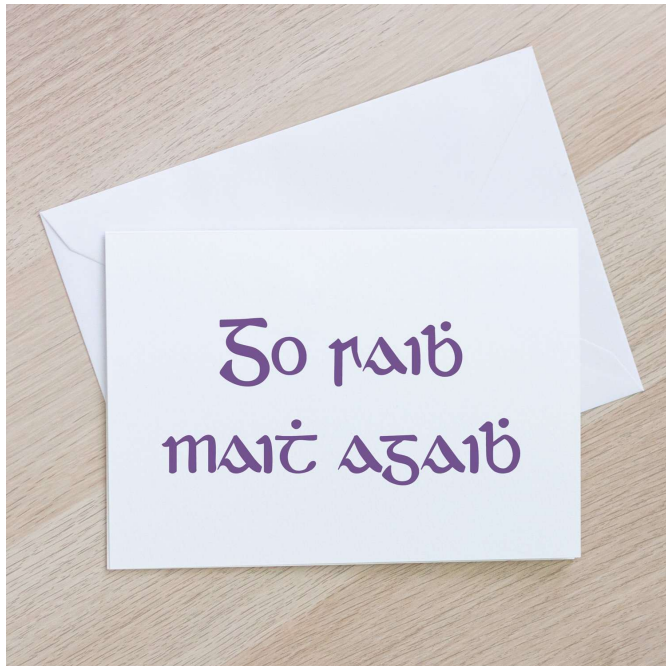
OCIMS Data Flows





Thanks... Questions, Comments...?

**Go raibh maith agaibh!!
Thank You!!**



Acknowledgments

HPSC Colleagues:

- Sarah Jackson (TB/STI Team)
- Mairead O'Hanlon (ARHCAI Team)
- Piaras O'Lorcain (VPD Team)
- Anthony Ortiz (GZV Team)
- Elaine Brabazon (RVU/OCIMS Team)
- Louise Cullen (OCIMS Team)
- Gareth Doherty (CIDR Team)
- Lisa Domegan (RVU Team)
- All HPSC staff

ECDC

Partner Laboratories:

- IMSRL
- IMRL (GRL/MRSA RL)
- VTEC/Campy/CDI RL
- NSSLRL, CPE RL
- NVRL
- Clinical Microbiology Laboratories

